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Switch system

Abstract

In a switch system, a voltage clamping element is connected to a semiconductor switch in parallel. An active clamping circuit is connected between a control terminal and a first main terminal of the semiconductor switch. The active clamping circuit includes a first diode, a second diode, and a control switch. A second anode of the second diode is connected to a first anode of the first diode. The control switch is connected between the first anode of the first diode and the control terminal of the semiconductor switch. A second control unit controls the control switch. A breakdown voltage of the first diode is smaller than a clamp voltage of the voltage clamping element.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6087877	12/1999	Gonda et al.	N/A	N/A
2005/0168264	12/2004	Fukami	327/535	H03K 17/0822
2005/0189965	12/2004	Tihanyi	327/108	H03K 17/302
2007/0008747	12/2006	Soldano et al.	N/A	N/A
2009/0052211	12/2008	Asai	363/56.05	H02M 7/5387
2009/0296291	12/2008	Volke	361/56	H03K 17/0822
2010/0033230	12/2009	Lueders et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102018112509	12/2020	DE	H01L 23/60
H11-032429	12/1998	JP	N/A
2002151989	12/2001	JP	H03K 17/0826
2005-223399	12/2004	JP	N/A
2007-028894	12/2006	JP	N/A
2007-166734	12/2006	JP	N/A
2008-539625	12/2007	JP	N/A
2012-004979	12/2011	JP	N/A
2017-022510	12/2016	JP	N/A

OTHER PUBLICATIONS

International Search Report dated Jun. 14, 2022 issued in International Patent Application No. PCT/JP2022/013914, with English translation. cited by applicant

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Background/Summary

(1) This application is the U.S. National Phase under 35 U.S.C. § 371 of International Patent Application No. PCT/JP2022/013914, filed on Mar. 24, 2022, which in turn claims the benefit of Japanese Patent Application No. 2021-089571, filed on May 27, 2021, the entire disclosures of which Applications are incorporated by reference herein.

TECHNICAL FIELD

(2) The present disclosure generally relates to a switch system, and more particularly relates to a switch system including a semiconductor switch.

BACKGROUND ART

(3) Patent Literature 1 discloses an active clamping circuit (switch system) including: a first switch element (semiconductor switch); a first diode causing breakdown due to an overvoltage across the first switch element; a first resistor for detecting a current flowing through the first diode; and a control circuit for controlling a current flowing through the first switch element by amplifying the voltage across the first resistor. The first switch element is implemented as an N-channel MOSFET.

(4) In the switch system of Patent Literature 1, if the energy generated in the inductance component of the line of a load circuit connected across the semiconductor switch exceeds the quantity that the semiconductor switch can withstand, then the characteristics of the semiconductor switch may deteriorate even when the clamp voltage is stabilized.

CITATION LIST

Patent Literature

(5) Patent Literature 1: JP 2012-4979 A

SUMMARY OF INVENTION

(6) It is therefore an object of the present disclosure to provide a switch system with the ability to reduce a surge voltage applied to a semiconductor switch when the semiconductor switch turns off.

(7) A switch system according to an aspect of the present disclosure includes a semiconductor switch, a voltage clamping element, an active clamping circuit, a first control unit, and a second control unit. The semiconductor switch has a control terminal, a first main terminal, and a second main terminal. The voltage clamping element is connected to the semiconductor switch in parallel. The active clamping circuit is connected between the control terminal and the first main terminal of the semiconductor switch. The first control unit controls the semiconductor switch. The second control unit controls the active clamping circuit. The active clamping circuit includes a first diode, a second diode, and a control switch. The first diode has a first anode and a first cathode. The first diode causes breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch. The second diode has a second anode and a second cathode. The second anode of the second diode is connected to the first anode of the first diode. The control switch is connected between the first anode of the first diode and the control terminal of the semiconductor switch. The second control unit is configured to control the control switch. A breakdown voltage of the first diode is smaller than a clamp voltage of the voltage clamping element.

(8) A switch system according to another aspect of the present disclosure includes a semiconductor switch, a voltage clamping element, an active clamping circuit, a first control unit, and a second control unit. The semiconductor switch has a control terminal, a first main terminal, and a second main terminal. The voltage clamping element is connected to the semiconductor switch in parallel. The active clamping circuit is connected between the control terminal and the first main terminal of the semiconductor switch. The first control unit controls the semiconductor switch. The second control unit controls the active clamping circuit. The active clamping circuit includes a first diode, a second diode, and a control switch. The first diode has a first anode and a first cathode. The first diode causes breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch. The second diode has a second anode and a second cathode. The second anode of the second diode is connected to the first anode of the first diode. The control switch is connected between the first anode of the first diode and the control terminal of the semiconductor switch. The control switch is a pnp transistor having a base, a collector, and an emitter. The emitter of the pnp transistor is connected to the second cathode of the second diode. The collector of the pnp transistor is connected to the control terminal of the semiconductor switch. The second control unit includes a first resistor, a second resistor, a field effect transistor, and a delay circuit. The first resistor is connected between the base and the emitter of the pnp transistor. The second resistor is connected to the base of the pnp transistor and connected to the first resistor in series. The field effect transistor has a gate, a drain, and a source. The field effect transistor is

connected between the second resistor and the second main terminal of the semiconductor switch. The delay circuit is connected between the first control unit and the gate of the field effect transistor. The delay circuit includes a gate resistor and a capacitor. The gate resistor is connected between the first control unit and the gate of the field effect transistor. The capacitor is connected between the gate and the source of the field effect transistor. A breakdown voltage of the first diode is smaller than a clamp voltage of the voltage clamping element.

(9) A switch system according to still another aspect of the present disclosure includes a semiconductor switch, a voltage clamping element, an active clamping circuit, a first control unit, and a second control unit. The semiconductor switch has a control terminal, a first main terminal, and a second main terminal. The voltage clamping element is connected to the semiconductor switch in parallel. The active clamping circuit is connected between the control terminal and the first main terminal of the semiconductor switch. The first control unit controls the semiconductor switch. The second control unit controls the active clamping circuit. The active clamping circuit includes a first diode, a second diode, and a control switch. The first diode has a first anode and a first cathode. The first diode causes breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch. The second diode has a second anode and a second cathode. The second cathode of the second diode is connected to the first cathode of the first diode. The control switch is connected between the first anode of the first diode and the control terminal of the semiconductor switch. The control switch is a field effect transistor having a gate, a drain, and a source. The drain of the field effect transistor is connected to the first anode of the first diode. The source of the field effect transistor is connected to the control terminal of the semiconductor switch. The second control unit includes a capacitor and a Zener diode. The capacitor has a first terminal and a second terminal. The first terminal of the capacitor is connected to the gate of the field effect transistor. The second terminal of the capacitor is connected to the first cathode of the first diode and the second cathode of the second diode. The Zener diode has a third anode and a third cathode. The third cathode of the Zener diode is connected to the first terminal of the capacitor and the gate of the field effect transistor. The third anode of the Zener diode is connected to the source of the field effect transistor.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a circuit diagram of a switch system according to a first embodiment;
- (2) FIG. 2 illustrates how the switch system operates;
- (3) FIG. 3 is a circuit diagram of a switch system according to a second embodiment;
- (4) FIG. 4 is a circuit diagram of a switch system according to a third embodiment;
- (5) FIG. 5 is a circuit diagram of a switch system according to a fourth embodiment;
- (6) FIG. 6 is a circuit diagram of a switch system according to a fifth embodiment;
- (7) FIG. 7 is a circuit diagram of a switch system according to a sixth embodiment;
- (8) FIG. 8 is a circuit diagram of a switch system according to a seventh embodiment;
- (9) FIG. 9 illustrates how the switch system operates;
- (10) FIG. 10 is a circuit diagram of a switch system according to an eighth embodiment;
- (11) FIG. 11 is a circuit diagram of a switch system according to a ninth embodiment;
- (12) FIG. 12 illustrates how the switch system operates;
- (13) FIG. 13 is a circuit diagram of a switch system according to a tenth embodiment;
- (14) FIG. 14 illustrates how the switch system operates;
- (15) FIG. 15 is a circuit diagram of a switch system according to an eleventh embodiment;
- (16) FIG. 16 illustrates how the switch system operates;
- (17) FIG. 17 is a circuit diagram of a switch system according to a twelfth embodiment;

- (18) FIG. **18** is a circuit diagram of a switch system according to a thirteenth embodiment;
- (19) FIG. **19** is a circuit diagram of a switch system according to a fourteenth embodiment;
- (20) FIG. **20** is a circuit diagram of a switch system according to a fifteenth embodiment;
- (21) FIG. **21** is a circuit diagram of a switch system according to a sixteenth embodiment;
- (22) FIG. **22** is a circuit diagram of a switch system according to a seventeenth embodiment;
- (23) FIG. **23** is a circuit diagram of a switch system according to an eighteenth embodiment;
- (24) FIG. **24** illustrates how the switch system operates; and
- (25) FIG. **25** is a circuit diagram of a switch system according to a nineteenth embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

(26) A switch system **20** according to a first embodiment will be described with reference to FIGS. **1** and **2**.

(1.1) Overview

(27) As shown in FIG. **1**, the switch system **20** includes a semiconductor switch **1**, a voltage clamping element **2**, an active clamping circuit **3**, a first control unit **4**, and a second control unit **5**. The semiconductor switch **1** has a control terminal **10**, a first main terminal **11**, and a second main terminal **12**. The voltage clamping element **2** is connected to the semiconductor switch **1** in parallel. The active clamping circuit **3** is connected between the control terminal **10** and the first main terminal **11** of the semiconductor switch **1**. The first control unit **4** controls the semiconductor switch **1**. The second control unit **5** controls the active clamping circuit **3**.

(28) In the switch system **20**, a load circuit **204** including a series circuit of a load **201** and a power supply **202** is connected between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**, for example. The switch system **20** further includes a first external terminal Tm1, to which the first main terminal **11** of the semiconductor switch **1** is connected, and a second external terminal Tm2, to which the second main terminal **12** of the semiconductor switch **1** is connected, for example. In the switch system **20**, the load circuit **204** is connected between the first external terminal Tm1 and the second external terminal Tm2, for example. The load **201** and the power supply **202** are not constituent elements of the switch system **20**. The power supply **202** may be a DC power supply, for example. The DC power supply may have an output voltage of 100 V to 400 V, for example.

(29) In the switch system **20**, the first external terminal Tm1 and the second external terminal Tm2 are terminals, through which a main current I1 flowing through the semiconductor switch **1** flows, when the semiconductor switch **1** is in on state (i.e., electrically conductive).

(30) (1.2) Respective Constituent Elements of Switch System

(31) (1.2.1) Semiconductor Switch

(32) The semiconductor switch **1** may be, for example, a junction field effect transistor Tr1.

(33) The junction field effect transistor Tr may be, for example, a GaN-based gate injection transistor (GIT). In this case, the junction field effect transistor Tr includes, for example, a substrate, a buffer layer, a first nitride semiconductor layer, a second nitride semiconductor layer, a source electrode, a gate electrode, a drain electrode, and a p-type layer. The buffer layer is formed on the substrate. The first nitride semiconductor layer is formed on the buffer layer. The second nitride semiconductor layer is formed on the first nitride semiconductor layer. The source electrode, the gate electrode, and the drain electrode are formed on the second nitride semiconductor layer. The p-type layer is interposed between the gate electrode and the second nitride semiconductor layer. In the junction field effect transistor Tr1, a diode structure is formed by the second nitride semiconductor layer and the p-type layer. The substrate may be a silicon substrate, for example. The buffer layer may be an undoped GaN layer, for example. The first nitride semiconductor layer may be, for example, an undoped GaN layer. The second nitride semiconductor layer may be, for example, an undoped AlGaN layer. The p-type layer may be, for example, a p-type AlGaN layer. Each of the buffer layer, the first nitride semiconductor layer, and the second nitride semiconductor

layer may include impurities such as Mg, H, Si, C, and O to be inevitably contained during their growing process by metal-organic vapor phase epitaxy (MOVPE), for example.

(34) The control terminal **10** of the semiconductor switch **1** is a gate terminal **G1** connected to the gate electrode of the junction field effect transistor **Tr1**. The first main terminal **11** of the semiconductor switch **1** is a drain terminal **D1** connected to the drain electrode of the junction field effect transistor **Tr1**. The second main terminal **12** of the semiconductor switch **1** is a source terminal **S1** connected to the source electrode of the junction field effect transistor **Tr1**.

(35) (1.2.2) Voltage Clamping Element

(36) The voltage clamping element **2** is connected to the semiconductor switch **1** in parallel as described above. The voltage clamping element **2** has an overvoltage protection function of restricting the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off to a fixed clamp voltage. That is to say, the voltage clamping element **2** has the function of restricting the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** to the clamp voltage when the semiconductor switch **1** turns off. The voltage clamping element **2** may be, for example, a Zener diode (such as a TVS diode). In the Zener diode serving as the voltage clamping element **2**, the anode thereof is connected to the second main terminal **12** of the semiconductor switch **1** and the cathode thereof is connected to the first main terminal **11** of the semiconductor switch **1**. The voltage clamping element **2** does not have to be a Zener diode but may also be varistor. The voltage clamping element **2** has the function of reducing, when a voltage equal to or larger than a predetermined voltage value is applied thereto, the chances of the voltage across the voltage clamping element **2** increasing to a voltage value larger than the predetermined voltage value. In the meantime, an electric current **I2** flows through the voltage clamping element **2**.

(37) The larger the current value of a current **I2**, flowing through the voltage clamping element **2** when the voltage clamping element **2** starts performing the clamping operation, is, the higher the clamp voltage of the voltage clamping element **2** is. That is why if the main current **I1** flowing through the semiconductor switch **1** is cut off, the clamp voltage of the voltage clamping element **2** varies according to the magnitude of the current value of the main current **I1** that flowed through the semiconductor switch **1** just before the main current **I1** is cut off.

(38) (1.2.3) Active Clamping Circuit

(39) The active clamping circuit **3** includes a first diode **31**, a second diode **32**, and a control switch **33**.

(40) The first diode **31** has a first anode and a first cathode. In the first diode **31**, the first cathode thereof is connected to the first main terminal **11** of the semiconductor switch **1**. The first diode **31** causes breakdown upon application of voltage between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1**. The breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. The first diode **31** may be, for example, a Zener diode.

(41) The second diode **32** has a second anode and a second cathode. In the second diode **32**, the second anode thereof is connected to the first anode of the first diode **31**. The second cathode of the second diode **32** is connected to the control terminal **10** of the semiconductor switch **1** via the control switch **33**.

(42) The control switch **33** is connected between the second cathode of the second diode **32** and the control terminal **10** of the semiconductor switch **1**. Thus, the control switch **33** is connected between the first anode of the first diode **31** and the control terminal **10** of the semiconductor switch **1**.

(43) The control switch **33** may be, for example, a field effect transistor **Q1** having a gate, a drain, and a source. In the control switch **33**, the drain of the field effect transistor **Q1** is connected to the second cathode of the second diode **32**, the source of the field effect transistor **Q1** is connected to the control terminal **10** of the semiconductor switch **1**, and the gate of the field effect transistor **Q1**

is connected to the second control unit **5**. In this embodiment, the field effect transistor **Q1** is a junction field effect transistor. However, this is only an example and should not be construed as limiting. Alternatively, the field effect transistor **Q1** may also be, for example, a metal-oxide semiconductor field effect transistor (MOSFET).

(44) (1.2.4) First Control Unit

(45) The first control unit **4** controls the semiconductor switch **1**. The first control unit **4** is connected to the control terminal **10** of the semiconductor switch **1**. More specifically, the first control unit **4** is connected between the control terminal **10** and second main terminal **12** of the semiconductor switch **1**. The first control unit **4** includes a first DC power supply (first drive power supply) and a first complementary metal-oxide semiconductor (CMOS) inverter. The first control unit **4** may output, as a first control signal for controlling the semiconductor switch **1**, a first on signal to turn the semiconductor switch **1** on (e.g., a voltage signal of 15 V) or a first off signal to turn the semiconductor switch **1** off (e.g., a voltage signal of 0 V). The first control unit **4** may change the voltage value of the first control signal within the range from 0 V to 15 V, for example.

(46) (1.2.5) Second Control Unit

(47) The second control unit **5** is configured to control the control switch **33**. The second control unit **5** is connected to the gate of the field effect transistor **Q1** serving as the control switch **33**.

(48) The second control unit **5** is connected between the gate of the field effect transistor **Q1** and the second main terminal **12** of the semiconductor switch **1**. The second control unit **5** includes a second DC power supply (second drive power supply) and a second complementary metal-oxide semiconductor (CMOS) inverter. The second control unit **5** may output, as a second control signal for controlling the control switch **33**, a second on signal to turn the field effect transistor **Q1** on (e.g., a voltage signal of 15 V) or a second off signal to turn the field effect transistor **Q1** off (e.g., a voltage signal of 0 V). The second control unit **5** may change the voltage value of the second control signal within the range from 0 V to 15 V, for example.

(49) (1.3) Exemplary Operation of Switch System

(50) Next, an exemplary operation of the switch system **20** will be described with reference to FIG. 2. The waveform labeled as “semiconductor switch (Tr1)” indicates how the first control signal supplied to the control terminal **10** of the semiconductor switch **1** (junction field effect transistor Tr1) changes with time. Also, in FIG. 2, the waveform labeled as “control switch (Q1)” indicates how the second control signal supplied to the control switch **33** (field effect transistor **Q1**) changes with time. Furthermore, in FIG. 2, “Vgs” denotes a voltage between the gate terminal **G1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor Tr1. Thus, “Vgs” denotes the voltage between the control terminal **10** and second main terminal **12** of the semiconductor switch **1**. Furthermore, in FIG. 2, “Vds” denotes a voltage between the drain terminal **D1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor Tr1. Thus, “Vds” denotes the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. The voltage clamping element **2** is connected to the semiconductor switch **1** in parallel. Thus, the voltage across the voltage clamping element **2** is the same as the voltage Vds between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(51) FIG. 2 illustrates how the switch system **20** operates, for example, in a situation where a short-circuit failure occurs to the load circuit **204** at a time **t1** to cause the main current **I1** of the semiconductor switch **1** to start to increase.

(52) When the second control signal output from the second control unit **5** changes from an off signal (second off signal) into an on signal (second on signal) at a time **t2**, the control switch **33** of the active clamping circuit **3** is turned on.

(53) Thereafter, when the main current **I1** reaches a threshold value at a time **t3** to cause the first control signal output from the first control unit **4** to change from an on signal (first on signal) into an off signal (first off signal), for example, the semiconductor switch **1** is turned off. In this switch

system **20**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, the first diode **31** causes breakdown earlier than the voltage clamping element **2** and the current flowing through the first diode **31** flows into the control terminal **10** of the semiconductor switch **1** via the voltage clamping element **2** and the control switch **33**. Consequently, the semiconductor switch **1** makes a transition from an off state (non-conductive state) into a semiconductive state where a main current **I1** flows to make the voltage value of the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** approximately equal to the clamp voltage of the first diode **31**. As a result, the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3** and part of the energy stored in the inductance component of a line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. The main current **I1** flowing through the semiconductor switch **1** starts to decrease at the time t_3 . The semiconductive state is an intermediate state between the non-conductive state (off state) and the conductive state (on state). The semiconductive state may be, for example, a state where a voltage of 200 V is applied to the semiconductor switch **1** when the power supply **202** as a DC power supply has an output voltage of 400 V and the active clamping circuit **3** has a clamp voltage of 200 V. That is to say, the semiconductive state herein refers to a state where the product of the resistance value of the semiconductor switch **1** and a leakage current flowing through the semiconductor switch **1** (i.e., the main current **I1**) is 200 V.

(54) At a time t_4 , when a predetermined period T_O passes since the time t_3 , the second control signal output from the second control unit **5** changes from an on signal (second on signal) into an off signal (second off signal). In response, the semiconductor switch **1** turns off, a current **I2** starts to flow through the voltage clamping element **2**, the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped at the clamp voltage of the voltage clamping element **2**, and the absolute value of a current variation ratio ($-dI_1/dt$) of the main current **I1** of the semiconductor switch **1** increases to cause the main current **I1** to decrease steeply to zero. Then, the current **I2** of the voltage clamping element **2** starts to decrease.

(55) When the current **I2** of the voltage clamping element **2** goes zero at a time t_5 , the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** becomes a constant voltage lower than the clamp voltage.

(56) The switch system **20** may make the current value of the main current **I1** flowing through the semiconductor switch **1** at the time t_4 smaller than the current value of the main current **I1** flowing through the semiconductor switch **1** at the time t_3 . This allows the switch system **20** to reduce the clamp voltage of the voltage clamping element **2** and thereby lower the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(57) In the switch system **20**, while the first control unit **4** is outputting the first on signal to turn the semiconductor switch **1** on before outputting the first off signal to turn the semiconductor switch **1** off, the second control unit **5** outputs a second on signal to turn the control switch **33** on. Also, in this switch system **20**, when the predetermined period T_O has passed since the output of the first off signal from the first control unit **4**, the second control unit **5** outputs a second off signal that turns the control switch **33** off. The switch system **20** includes a timing control unit for controlling the first control unit **4** and the second control unit **5** when any error occurs to the load circuit **204**, for example. The timing control unit controls the first control unit **4** and the second control unit **5** when an error occurs to the load circuit **204** to make the main current **I1** flowing through the semiconductor switch **1** larger than a threshold value.

(58) (1.4) Advantages

(59) A switch system **20** according to the first embodiment includes a semiconductor switch **1**, a voltage clamping element **2**, an active clamping circuit **3**, a first control unit **4**, and a second control unit **5**. The semiconductor switch **1** has a control terminal **10**, a first main terminal **11**, and a second main terminal **12**. The voltage clamping element **2** is connected to the semiconductor switch **1** in

parallel. The active clamping circuit 3 is connected between the control terminal 10 and the first main terminal 11 of the semiconductor switch 1. The first control unit 4 controls the semiconductor switch 1. The second control unit 5 controls the active clamping circuit 3. The active clamping circuit 3 includes a first diode 31, a second diode 32, and a control switch 33. The first diode 31 has a first anode and a first cathode. The first diode 31 causes breakdown upon application of voltage between the first main terminal 11 and the second main terminal 12 of the semiconductor switch 1. The second diode 32 has a second anode and a second cathode. The second anode of the second diode 32 is connected to the first anode of the first diode 31. The control switch 33 is connected between the first anode of the first diode 31 and the control terminal 10 of the semiconductor switch 1. The second control unit 5 is configured to control the control switch 33. A breakdown voltage of the first diode 31 is smaller than a clamp voltage of the voltage clamping element 2.

(60) The switch system 20 according to the first embodiment may reduce a surge voltage applied to the semiconductor switch 1 when the semiconductor switch 1 turns off. More specifically, the switch system 20 according to the first embodiment may reduce the current value of the main current I1 flowing through the semiconductor switch 1 and thereby reduce the clamp voltage when the voltage clamping element 2 clamps the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1. This allows the switch system 20 to reduce the surge voltage applied between the first main terminal 11 and the second main terminal 12 of the semiconductor switch 1 when the semiconductor switch 1 turns off. Consequently, the switch system 20 may reduce deterioration in the characteristics of the semiconductor switch 1 and the voltage clamping element 2 (e.g., may reduce the chances of the semiconductor switch 1 and the voltage clamping element 2 coming to have a shorter life). In addition, the switch system 20 may reduce the clamp voltage, and therefore, may also reduce deterioration in the characteristics of the load 201 of the load circuit 204.

Second Embodiment

(61) Next, a switch system 20A according to a second embodiment will be described with reference to FIG. 3.

(62) The switch system 20A according to the second embodiment is substantially the same as the switch system 20 according to the first embodiment (refer to FIG. 1) but includes an active clamping circuit 3A and a second control unit 5A instead of the active clamping circuit 3 and the second control unit 5, respectively, which is a difference from the switch system 20 according to the first embodiment. In the following description, any constituent element of the switch system 20A according to this second embodiment, having the same function as a counterpart of the switch system 20 according to the first embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(63) The active clamping circuit 3A, as well as the active clamping circuit 3, includes a first diode 31, a second diode 32, and a control switch 33. In the following description, a field effect transistor Q1 serving as the control switch 33 will be hereinafter sometimes referred to as a "first field effect transistor Q1" and the gate, drain, and source of the field effect transistor Q1 will be hereinafter sometimes referred to as a first gate, a first drain, and a first source, respectively.

(64) The active clamping circuit 3A further includes a second field effect transistor Q2. The second field effect transistor Q2 has a second gate, a second drain, and a second source. The second field effect transistor Q2 is connected between the first gate of the first field effect transistor Q1 and the second main terminal 12 of the semiconductor switch 1. In the second field effect transistor Q2, the second drain thereof is connected to the first gate of the first field effect transistor Q1, the second source thereof is connected to the second main terminal 12 of the semiconductor switch 1, and the second gate thereof is connected to the second control unit 5. In this embodiment, the second field effect transistor Q2 is a junction field effect transistor. However, this is only an example and should not be construed as limiting. Alternatively, the second field effect transistor Q2 may also be a MOSFET.

(65) The second control unit 5A controls the first field effect transistor Q1 and the second field effect transistor Q2.

(66) The switch system 20A turns the first field effect transistor Q1 off by making the second control unit 5A turn the second field effect transistor Q2 on. When a predetermined period has passed since the turn on of the first field effect transistor Q1, the second control unit 5A turns the second field effect transistor Q2 on, thereby turning the first field effect transistor Q1 off. This allows the second control unit 5A to keep the control switch 33 of the active clamping circuit 3A on only for the predetermined period.

(67) In the switch system 20A, as well as in the switch system 20, the breakdown voltage of the first diode 31 is smaller than the clamp voltage of the voltage clamping element 2. Thus, in the switch system 20A, when the control switch 33 (first field effect transistor Q1) is turned on to turn the semiconductor switch 1 off, the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3A. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. In the switch system 20A, when the control switch 33 of the active clamping circuit 3A turns from on to off, the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the voltage clamping element 2.

(68) The switch system 20A according to the second embodiment, as well as the switch system 20 according to the first embodiment, may also reduce the surge voltage applied to the semiconductor switch 1 when the semiconductor switch 1 turns off.

Third Embodiment

(69) Next, a switch system 20B according to a third embodiment will be described with reference to FIG. 4.

(70) The switch system 20B according to the third embodiment is substantially the same as the switch system 20A according to the second embodiment (refer to FIG. 3) but further includes a delay circuit 6, which is a difference from the switch system 20A according to the second embodiment. In the following description, any constituent element of the switch system 20B according to this third embodiment, having the same function as a counterpart of the switch system 20A according to the second embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(71) The delay circuit 6 is connected between a signal path that connects the second control unit 5A to the first gate of the first field effect transistor Q1 and the second gate of the second field effect transistor Q2. Thus, in this switch system 20B, the second gate of the second field effect transistor Q2 is connected to the second control unit 5A via the delay circuit 6.

(72) In this switch system 20B, the first field effect transistor Q1 and the second field effect transistor Q2 turn on and off in response to the second control signal supplied from the second control unit 5A. The switching operation of the second field effect transistor Q2 trails by a predetermined period (i.e., the time delay introduced by the delay circuit 6) behind the switching operation of the first field effect transistor Q1. This causes the active clamping circuit 3A to turn off when the predetermined period has passed since the turn on of the first field effect transistor Q1 serving as the control switch 33.

(73) In the switch system 20B, as well as in the switch system 20A, the breakdown voltage of the first diode 31 is smaller than the clamp voltage of the voltage clamping element 2. Thus, in the switch system 20B, when the control switch 33 (first field effect transistor Q1) is turned on to turn the semiconductor switch 1 off, the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3A. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. In the switch system 20B, when the control switch 33 of the active clamping circuit 3A turns from on to off, the voltage between the first main terminal

11 and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**.

(74) The switch system **20B** according to the third embodiment, as well as the switch system **20A** according to the second embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Fourth Embodiment

(75) Next, a switch system **20C** according to a fourth embodiment will be described with reference to FIG. 5.

(76) The switch system **20C** according to the fourth embodiment is substantially the same as the switch system **20A** according to the second embodiment (refer to FIG. 3) but further includes a NOT circuit **7**, which is a difference from the switch system **20A** according to the second embodiment. In the following description, any constituent element of the switch system **20C** according to this fourth embodiment, having the same function as a counterpart of the switch system **20A** according to the second embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(77) The NOT circuit **7** is connected between a signal path that connects the first control unit **4** to the control terminal **10** of the semiconductor switch **1** and the first gate of the first field effect transistor **Q1**. The second control unit **5A** is connected to the second gate of the second field effect transistor **Q2**.

(78) When the first control signal output from the first control unit **4** to control the semiconductor switch **1** changes from an on signal into an off signal, the NOT circuit **7** outputs an on signal to the first field effect transistor **Q1**. Thus, in the switch system **20C**, when the semiconductor switch **1** turns off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** increases to energize the active clamping circuit **3A**. When a predetermined period has passed, the second control unit **5A** will turn on the second field effect transistor **Q2** in the off state. As a result, in the switch system **20C**, the first field effect transistor **Q1** turns off. Then, in the switch system **20C**, the semiconductor switch **1** turns completely off. Consequently, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** further increases, thus causing the voltage clamping element **2** to start performing the clamping operation.

(79) In the switch system **20C**, as well as in the switch system **20A**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, in the switch system **20C**, when the control switch **33** (first field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3A**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20C**, when the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**.

(80) The switch system **20C** according to the fourth embodiment, as well as the switch system **20A** according to the second embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Fifth Embodiment

(81) Next, a switch system **20D** according to a fifth embodiment will be described with reference to FIG. 6.

(82) The switch system **20D** according to the fifth embodiment is substantially the same as the switch system **20C** according to the fourth embodiment (refer to FIG. 5) but includes a second control unit **5D** instead of the second control unit **5A**, which is a difference from the switch system **20C** according to the fourth embodiment. In the following description, any constituent element of

the switch system **20D** according to this fifth embodiment, having the same function as a counterpart of the switch system **20C** according to the fourth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(83) In the switch system **20D**, the second control unit **5D** includes a delay circuit **52**. The delay circuit **52** is connected between a signal path that connects the output terminal of the NOT circuit **7** to the first field effect transistor **Q1** and the second gate of the second field effect transistor **Q2**. The delay circuit **52** outputs a signal produced by delaying the input signal for the predetermined period. The delay circuit **52** may be, but does not have to be, an analog delay circuit, for example. Alternatively, the delay circuit **52** may also be timer IC or implemented as an integrated circuit (IC) including the first control unit **4**.

(84) The signal supplied to the delay circuit **52** of the second control unit **5D** is produced by having the first control signal output from the first control unit **4** inverted by the NOT circuit **7**. An output signal of the second control unit **5D** is produced by having the signal supplied to the delay circuit **52** delayed for the predetermined period by the delay circuit **52**. Thus, the timing when the second field effect transistor **Q2** is switched is delayed by the predetermined period from the timing when the field effect transistor **Q1** is switched.

(85) In the switch system **20D**, when the first control signal changes from a first on signal into a first off signal, the second control signal changes from a second off signal into a second on signal. As a result, while the semiconductor switch **1** turns off, the first field effect transistor **Q1** turns from off to on. When the predetermined period has passed since the turn on of the first field effect transistor **Q1**, the second field effect transistor **Q2** will turn from off to on. When the second field effect transistor **Q2** turns on, the first field effect transistor **Q1** turns from on to off.

(86) In the switch system **20D**, as well as in the switch system **20C**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, in the switch system **20D**, when the control switch **33** (first field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3A**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20D**, when the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**.

(87) The switch system **20D** according to the fifth embodiment, as well as the switch system **20C** according to the fourth embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Sixth Embodiment

(88) Next, a switch system **20E** according to a sixth embodiment will be described with reference to FIG. 7.

(89) The switch system **20E** according to the sixth embodiment is substantially the same as the switch system **20D** according to the fifth embodiment (refer to FIG. 6) but further includes a first drive circuit **8**, which is a difference from the switch system **20D** according to the fifth embodiment. In addition, the switch system **20E** according to the sixth embodiment includes a second control unit **5E** instead of the second control unit **5D** of the switch system **20D** according to the fifth embodiment, which is another difference from the switch system **20D** according to the fifth embodiment. In the following description, any constituent element of the switch system **20E** according to this sixth embodiment, having the same function as a counterpart of the switch system **20D** according to the fifth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(90) The first drive circuit **8** includes a first gate resistor **Rg1**. The first gate resistor **Rg1** is connected between the output terminal of the NOT circuit **7** and the first gate of the first field effect

transistor Q1. In the second control unit 5E, the delay circuit 52 is implemented as a second drive circuit 9 including a second gate resistor Rg2 and a capacitor C2. The second gate resistor Rg2 is connected between the output terminal of the NOT circuit 7 and the second gate of the second field effect transistor Q2. The capacitor C2 is connected between the second gate and second source of the second field effect transistor Q2.

(91) In this switch system 20E, when the control signal supplied from the first control unit 4 changes from a first on signal into a first off signal while the semiconductor switch 1 is being turned off, the first field effect transistor Q1 turns on. In the predetermined period, the second field effect transistor Q2 will turn on as well. In the active clamping circuit 3A, when the second field effect transistor Q2 turns on, the first field effect transistor Q1 turns off.

(92) In the switch system 20E, as well as in the switch system 20D, the breakdown voltage of the first diode 31 is smaller than the clamp voltage of the voltage clamping element 2. Thus, in the switch system 20E, when the control switch 33 (first field effect transistor Q1) is turned on to turn the semiconductor switch 1 off, the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3A. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. In the switch system 20E, when the control switch 33 of the active clamping circuit 3A turns from on to off, the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the voltage clamping element 2.

(93) The switch system 20E according to the sixth embodiment, as well as the switch system 20D according to the fifth embodiment, may also reduce the surge voltage applied to the semiconductor switch 1 when the semiconductor switch 1 turns off.

(94) In addition, in the switch system 20E according to the sixth embodiment, the delay circuit 52 may be implemented as a combination of the second gate resistor Rg2 and the capacitor C2, thus contributing to cutting down the cost.

Seventh Embodiment

(95) Next, a switch system 20F according to a seventh embodiment will be described with reference to FIGS. 8 and 9.

(96) The switch system 20F according to the seventh embodiment is substantially the same as the switch system 20E according to the sixth embodiment (refer to FIG. 7) but further includes a third drive circuit 13, which is a difference from the switch system 20E according to the sixth embodiment. In addition, the switch system 20F according to the seventh embodiment includes a NOT circuit 7F instead of the NOT circuit 7 of the switch system 20E according to the sixth embodiment, which is another difference from the switch system 20E according to the sixth embodiment. In the following description, any constituent element of the switch system 20F according to this seventh embodiment, having the same function as a counterpart of the switch system 20E according to the sixth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(97) The third drive circuit 13 is connected between the first control unit 4 and the control terminal 10 of the semiconductor switch 1. The third drive circuit 13 includes a third gate resistor Rg3 connected between the first control unit 4 and the control terminal 10 of the semiconductor switch 1.

(98) The NOT circuit 7F includes a power supply 71, a resistor 72, a third field effect transistor Q3, and a fourth drive circuit 74. The power supply 71 is a DC power supply. The resistor 72 is connected to the power supply 71 in series. The third field effect transistor Q3 has a third gate, a third drain, and a third source. The third field effect transistor Q3 is connected to the resistor 72 in series. In the third field effect transistor Q3, the third drain thereof is connected to the resistor 72 and the third source thereof is connected to the second source of the second field effect transistor Q2 and the second main terminal 12 of the semiconductor switch 1. In this embodiment, the third

field effect transistor Q3 is a junction field effect transistor. However, this is only an example and should not be construed as limiting. Alternatively, the third field effect transistor Q3 may also be a MOSFET, for example. The fourth drive circuit 74 is connected between the first control unit 4 and the third gate of the third field effect transistor Q3. The fourth drive circuit 74 includes a gate resistor Rg4 (fourth gate resistor Rg4) connected between the first control unit 4 and the third gate of the third field effect transistor Q3.

(99) In the switch system 20F according to the seventh embodiment, when the control signal output from the first control unit 4 changes from a first on signal into a first off signal to turn the semiconductor switch 1 off, the time it takes to turn the first field effect transistor Q1 on is shorter than the time it takes for the semiconductor switch 1 to start turning off. In addition, in the switch system 20F, when a predetermined period has passed since the turn on of the first field effect transistor Q1, the second field effect transistor Q2 turns on. In the switch system 20F, when the second field effect transistor Q2 turns on, the first field effect transistor Q1 turns off.

(100) Next, an exemplary operation of the switch system 20F will be described with reference to FIG. 9. In FIG. 9, the waveform labeled as “semiconductor switch (Tr1)” indicates how the first control signal supplied to the control terminal 10 of the semiconductor switch 1 (junction field effect transistor Tr1) changes with time. Also, in FIG. 9, the waveform labeled as “control switch (Q1)” indicates how the second control signal supplied to the control switch 33 (first field effect transistor Q1) changes with time. In FIG. 9, the waveform labeled as “second FET (Q2)” indicates how a third control signal supplied to the second gate of the second field effect transistor Q2 changes with time. In FIG. 9, the waveform labeled as “third FET (Q3)” indicates how a control signal supplied to the third gate of the third field effect transistor Q3 changes with time.

Furthermore, in FIG. 9, “Vgs” denotes a voltage between the gate terminal G1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vgs” denotes the voltage between the control terminal 10 and second main terminal 12 of the semiconductor switch 1. Furthermore, in FIG. 9, “Vds” denotes a voltage between the drain terminal D1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vds” denotes the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1. The voltage clamping element 2 is connected to the semiconductor switch 1 in parallel. Therefore, the voltage across the voltage clamping element 2 is the same as the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1.

(101) FIG. 9 illustrates how the switch system 20F operates, for example, in a situation where a short-circuit failure occurs to the load circuit 204 at a time t1 to cause the main current I1 of the semiconductor switch 1 to start to increase.

(102) At a time t12, the first control signal changes from a first on signal into a first off signal, the second control signal changes from a second off signal into a second on signal, and the control signal for the third field effect transistor Q3 changes from an on signal into an off signal. This causes not only the semiconductor switch 1 to start turning off but also the control switch 33 (first field effect transistor Q1) of the active clamping circuit 3 to turn on. When the control switch 33 of the active clamping circuit 3 turns on, the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. The main current I1 flowing through the semiconductor switch 1 starts to decrease at the time t12.

(103) At a time t13, when the predetermined period TO passes since the time t12, the third control signal for the second field effect transistor Q2 changes from an off signal into an on signal and the second control signal for the control switch 33 changes from the on signal (second on signal) into the off signal (second off signal). In response, a current I2 starts to flow through the voltage clamping element 2, the voltage Vds between the first main terminal 11 and second main terminal

12 of the semiconductor switch **1** is clamped at the clamp voltage of the voltage clamping element **2**, and the absolute value of a current variation ratio ($-dI_1/dt$) of the main current **I1** of the semiconductor switch **1** increases to cause the main current **I1** to decrease steeply to zero. Then, the current **I2** of the voltage clamping element **2** starts to decrease.

(104) When the current **I2** of the voltage clamping element **2** goes zero at a time **t14**, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** becomes a constant voltage lower than the clamp voltage.

(105) The switch system **20F** may make the current value of the main current **I1** flowing through the semiconductor switch **1** at the time **t13** smaller than the current value of the main current **I1** flowing through the semiconductor switch **1** at the time **t12**. This allows the switch system **20F** to reduce the clamp voltage of the voltage clamping element **2**.

(106) In the switch system **20F**, as well as in the switch system **20E**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, in the switch system **20F**, when the control switch **33** (first field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3A**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20F**, when the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**.

(107) The switch system **20F** according to the seventh embodiment, as well as the switch system **20E** according to the sixth embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Eighth Embodiment

(108) Next, a switch system **20G** according to an eighth embodiment will be described with reference to FIG. **10**.

(109) The switch system **20G** according to the eighth embodiment is substantially the same as the switch system **20F** according to the seventh embodiment (refer to FIG. **8**) but includes a third drive circuit **13G** instead of the third drive circuit **13** of the switch system **20F** according to the seventh embodiment, which is a difference from the switch system **20F** according to the seventh embodiment. In addition, the switch system **20G** according to the eighth embodiment includes a NOT circuit **7G** instead of the NOT circuit **7F** of the switch system **20F** according to the seventh embodiment, which is another difference from the switch system **20F** according to the seventh embodiment. The NOT circuit **7G** does not include the fourth drive circuit **74** of the NOT circuit **7F**. Thus, the third gate of the third field effect transistor **Q3** is directly connected to the first control unit **4**. In the following description, any constituent element of the switch system **20G** according to this eighth embodiment, having the same function as a counterpart of the switch system **20F** according to the seventh embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(110) The third drive circuit **13G** has a first path **131** to turn the semiconductor switch **1** on and a second path **132** to turn the semiconductor switch **1** off. As used herein, the first path **131** to turn the semiconductor switch **1** on refers to a charging path for charging the control terminal **10** of the semiconductor switch **1** with electric charges. The second path **132** to turn the semiconductor switch **1** off refers to a discharging path for discharging electric charges from the control terminal **10** of the semiconductor switch **1**.

(111) The first path **131** includes a third diode **133** and a third gate resistor **Rg3**. The third diode **133** has a third anode and a third cathode. The third cathode of the third diode **133** is connected to the control terminal **10** of the semiconductor switch **1**. The third anode of the third diode **133** is connected to the first control unit **4**. The third gate resistor **Rg3** is connected to the third diode **133**

in series. In the first path **131**, the third cathode of the third diode **133** is control terminal **10** of the semiconductor switch **1** via the third gate resistor **Rg3**. However, this is only an example and should not be construed as limiting. Alternatively, the third cathode of the third diode **133** may be directly connected to the control terminal **10** of the semiconductor switch **1** and the third anode of the third diode **133** may be connected to the first control unit **4** via the third gate resistor **Rg3**. As used herein, the expression “the third cathode of the third diode **133** is directly connected to the control terminal **10** of the semiconductor switch **1**” means that the third cathode of the third diode **133** is connected to the control terminal **10** of the semiconductor switch **1** not via any other circuit component (such as the third gate resistor **Rg3**).

(112) The second path **132** includes a resistor **134**. The resistor **134** is connected between the first control unit **4** and the control terminal **10** of the semiconductor switch **1**. More specifically, in the second path **132**, one end of the resistor **134** is connected to the first control unit **4** and the other end of the resistor **134** is connected to the control terminal **10** of the semiconductor switch **1**. In the switch system **20G**, the first field effect transistor **Q1** of the active clamping circuit **3A** turns on before the semiconductor switch **1** turns off. The resistance value of the resistor **134** on the second path **132** is smaller than the resistance value of the third gate resistor **Rg3** on the first path **131**. This allows the switch system **20G** to reduce, when the junction field effect transistor **Tr** serving as the semiconductor switch **1** is a GaN-based GIT, for example, the effect of the resistor **134** on the turn-on characteristic when the semiconductor switch **1** is turned on. In addition, in the switch system **20G**, when the semiconductor switch **1** is turned off, not the first path **131** but the second path **132** is used, thus preventing the semiconductor switch **1** from turning off before the control switch **33** of the active clamping circuit **3A** turns on.

(113) In the switch system **20G**, as well as in the switch system **20F**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, in the switch system **20G**, when the control switch **33** (first field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3A**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20G**, when the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**.

(114) The switch system **20G** according to the eighth embodiment, as well as the switch system **20F** according to the seventh embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Ninth Embodiment

(115) Next, a switch system **20H** according to a ninth embodiment will be described with reference to FIGS. **11** and **12**.

(116) The switch system **20H** according to the ninth embodiment is substantially the same as the switch system **20** according to the first embodiment (refer to FIG. **1**) but includes an active clamping circuit **3H** and a second control unit **5H** instead of the active clamping circuit **3** and the second control unit **5** of the switch system **20** according to the first embodiment, which is a difference from the switch system **20** according to the first embodiment. In the following description, any constituent element of the switch system **20H** according to this ninth embodiment, having the same function as a counterpart of the switch system **20** according to the first embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(117) The active clamping circuit **3H** is connected between the control terminal **10** and first main terminal **11** of the semiconductor switch **1**. The active clamping circuit **3H** includes a control switch **33H** instead of the control switch **33** of the active clamping circuit **3** of the switch system **20**

according to the first embodiment. The control switch 33H is connected between the first anode of the first diode 31 and the control terminal 10 of the semiconductor switch 1. The control switch 33H is a pnp transistor Qp1 having a base, a collector, and an emitter. The emitter of the pnp transistor Qp1 is connected to the second cathode of the second diode 32. The collector of the pnp transistor Qp1 is connected to the control terminal 10 of the semiconductor switch 1. Thus, the collector of the pnp transistor Qp1 is connected to the signal path between the first control unit 4 and the control terminal 10 of the semiconductor switch 1.

(118) The second control unit 5H controls the active clamping circuit 3H. The second control unit 5H includes a first resistor R1, a second resistor R2, a field effect transistor Q4, and a delay circuit 54.

(119) The first resistor R1 is connected between the base and the emitter of the pnp transistor Qp1. Thus, the first resistor R1 is connected to the second cathode of the second diode 32. The second resistor R2 is connected to the base of the pnp transistor Qp1 and connected to the first resistor R1 in series.

(120) The field effect transistor Q4 has a gate, a drain, and a source. The field effect transistor Q4 is connected between the second resistor R2 and the second main terminal 12 of the semiconductor switch 1. In this embodiment, the field effect transistor Q4 is implemented as a junction field effect transistor. However, this is only an example and should not be construed as limiting. Alternatively, the field effect transistor Q4 may also be a MOSFET.

(121) The delay circuit 54 is connected between the first control unit 4 and the gate of the field effect transistor Q4. The delay circuit 54 includes a gate resistor Rg4 and a capacitor C4. The gate resistor Rg4 is connected between the first control unit 4 and the gate of the field effect transistor Q4. The capacitor C4 is connected between the gate and the source of the field effect transistor Q4. The delay circuit 54 delays the first control signal, supplied from the first control unit 4 to the semiconductor switch 1, for a predetermined period and provides the first control signal thus delayed to the gate of the field effect transistor Q4 of the active clamping circuit 3H.

(122) In the switch system 20H, as well as in the switch system 20, the breakdown voltage of the first diode 31 is also smaller than the clamp voltage of the voltage clamping element 2.

(123) Next, an exemplary operation of the switch system 20H will be described with reference to FIG. 12. In FIG. 12, the waveform labeled as “semiconductor switch (Tr1)” indicates how the first control signal supplied to the control terminal 10 of the semiconductor switch 1 (junction field effect transistor Tr1) changes with time. Also, in FIG. 12, the waveform labeled as “control switch (Qp1)” indicates how the state of the control switch 33H (pnp transistor Qp1) changes with time. In FIG. 12, the waveform labeled as “Q4” indicates how the control signal supplied to the gate of the field effect transistor Q4 changes with time. Furthermore, in FIG. 12, “Vgs” denotes a voltage between the gate terminal G1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vgs” denotes the voltage between the control terminal 10 and second main terminal 12 of the semiconductor switch 1. Furthermore, in FIG. 12, “Vds” denotes a voltage between the drain terminal D1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vds” denotes the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1. The voltage clamping element 2 is connected to the semiconductor switch 1 in parallel. Thus, the voltage across the voltage clamping element 2 is the same as the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1.

(124) FIG. 12 illustrates how the switch system 20H operates, for example, in a situation where a short-circuit failure occurs to the load circuit 204 at a time t21 to cause the main current I1 of the semiconductor switch 1 to start to increase.

(125) At a time t22, the first control signal changes from a first on signal into a first off signal. This causes the semiconductor switch 1 to start turning off. The field effect transistor Q4 is caused by the delay circuit 54 to start turning off later than the semiconductor switch 1. Thus, at the time t22,

the field effect transistor Q4 is still in on state. As the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 increases at the time t22, the first diode 31 causes breakdown to allow a current to flow through the first resistor R1, the second resistor R2, and the field effect transistor Q4. As a result, the pnp transistor Qp1 turns on. When the control switch 33 of the active clamping circuit 3H turns on, the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3H. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. The main current I1 flowing through the semiconductor switch 1 starts to decrease at the time t22.

(126) At a time t23, when the predetermined period TO passes since the time t22, the control signal for the field effect transistor Q4 changes from an on signal into an off signal to cause the control switch 33H to turn off. In response, in the switch system 20H, a current I2 starts to flow through the voltage clamping element 2, the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped at the clamp voltage of the voltage clamping element 2, and the absolute value of a current variation ratio ($-dI1/dt$) of the main current I1 of the semiconductor switch 1 increases to cause the main current I1 to decrease steeply to zero. Then, the current I2 of the voltage clamping element 2 starts to decrease.

(127) When the current I2 of the voltage clamping element 2 goes zero at a time t24, the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 becomes a constant voltage lower than the clamp voltage.

(128) The switch system 20F may make the current value of the main current I1 flowing through the semiconductor switch 1 at the time t23 smaller than the current value of the main current I1 flowing through the semiconductor switch 1 at the time t22. This allows the switch system 20H to reduce the clamp voltage of the voltage clamping element 2.

(129) A switch system 20H according to the ninth embodiment described above includes a semiconductor switch 1, a voltage clamping element 2, an active clamping circuit 3H, a first control unit 4, and a second control unit 5H. The semiconductor switch 1 has a control terminal 10, a first main terminal 11, and a second main terminal 12. The voltage clamping element 2 is connected to the semiconductor switch 1 in parallel. The active clamping circuit 3H is connected between the control terminal 10 and the first main terminal 11 of the semiconductor switch 1. The first control unit 4 controls the semiconductor switch 1. The second control unit 5H controls the active clamping circuit 3H. The active clamping circuit 3H includes a first diode 31, a second diode 32, and a control switch 33H. The first diode 31 has a first anode and a first cathode. The first diode 31 causes breakdown upon application of voltage between the first main terminal 11 and the second main terminal 12 of the semiconductor switch 1. The second diode 32 has a second anode and a second cathode. The second anode of the second diode 32 is connected to the first anode of the first diode 31. The control switch 33H is connected between the first anode of the first diode 31 and the control terminal 10 of the semiconductor switch 1. The control switch 33H is a pnp transistor Qp1 having a base, a collector, and an emitter. The emitter of the pnp transistor Qp1 is connected to the second cathode of the second diode 32. The collector of the pnp transistor Qp1 is connected to the control terminal 10 of the semiconductor switch 1. The second control unit 5H includes a first resistor R1, a second resistor R2, a field effect transistor Q4, and a delay circuit 54. The first resistor R1 is connected between the base and the emitter of the pnp transistor Qp1. The second resistor R2 is connected to the base of the pnp transistor Qp1 and connected to the first resistor R1 in series. The field effect transistor Q4 has a gate, a drain, and a source. The field effect transistor Q4 is connected between the second resistor R2 and the second main terminal 12 of the semiconductor switch 1. The delay circuit 54 is connected between the first control unit 4 and the gate of the field effect transistor Q4. The delay circuit 54 includes a gate resistor Rg4 and a capacitor C4. The gate resistor Rg4 is connected between the first control unit 4 and the gate of the field effect transistor Q4. The capacitor C4 is connected between the gate and the source of the

field effect transistor **Q4**. A breakdown voltage of the first diode **31** is smaller than a clamp voltage of the voltage clamping element **2**.

(130) The switch system **20H** according to the ninth embodiment may reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off. More specifically, the switch system **20H** according to the ninth embodiment may reduce the current value of the main current **I1** flowing through the semiconductor switch **1** when the voltage clamping element **2** clamps the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** (i.e., may reduce the clamp voltage). Consequently, the switch system **20H** may reduce deterioration in the characteristics of the semiconductor switch **1** (e.g., may reduce the chances of the semiconductor switch **1** coming to have a shorter life).

Tenth Embodiment

(131) Next, a switch system **20I** according to a tenth embodiment will be described with reference to FIGS. **13** and **14**.

(132) The switch system **20I** according to the tenth embodiment is substantially the same as the switch system **20** according to the first embodiment (refer to FIG. **1**) but includes an active clamping circuit **3I** and a second control unit **5I** instead of the active clamping circuit **3** and the second control unit **5** of the switch system **20** according to the first embodiment, which is a difference from the switch system **20** according to the first embodiment. In addition, the switch system **20I** according to the tenth embodiment includes a voltage clamping element **2I** instead of the voltage clamping element **2** of the switch system **20** according to the first embodiment, which is another difference from the switch system **20** according to the first embodiment. In the following description, any constituent element of the switch system **20I** according to this tenth embodiment, having the same function as a counterpart of the switch system **20** according to the first embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(133) The active clamping circuit **3I** is connected between the control terminal **10** and first main terminal **11** of the semiconductor switch **1**. The active clamping circuit **3I** includes a first diode **31**, a second diode **32**, and a control switch **33**. In the active clamping circuit **3I**, the second anode of the second diode **32** is connected to the first main terminal **11** of the semiconductor switch **1**, and the second cathode of the second diode **32** is connected to the control switch **33** via the first diode **31**. The second cathode of the second diode **32** is connected to the first cathode of the first diode **31**. The control switch **33** is connected between the first anode of the first diode **31** and the control terminal **10** of the semiconductor switch **1**. The first diode **31** causes breakdown upon application of voltage between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1**. The control switch **33** is a field effect transistor **Q1**. The drain of the field effect transistor **Q1** is connected to the first anode of the first diode **31**. The source of the field effect transistor **Q1** is connected to the control terminal **10** of the semiconductor switch **1**.

(134) The second control unit **5I** controls the active clamping circuit **3I**. The second control unit **5I** includes a capacitor **55** and a Zener diode **56**. The capacitor **55** has a first terminal and a second terminal. In the capacitor **55**, the first terminal thereof is connected to the gate of the field effect transistor **Q1**, and the second terminal thereof is connected to the first cathode of the first diode **31** and the second cathode of the second diode **32**. The Zener diode **56** has a third anode and a third cathode. In the Zener diode **56**, the third cathode thereof is connected to the first terminal of the capacitor **55** and the gate of the field effect transistor **Q1** and the third anode thereof is connected to the source of the field effect transistor **Q1**. Thus, the third anode of the Zener diode **56** is connected to the signal path between the first control unit **4** and the control terminal **10** of the semiconductor switch **1**.

(135) The voltage clamping element **2I** is connected to the semiconductor switch **1** in parallel. More specifically, the voltage clamping element **2I** is connected between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1**. The voltage clamping element **2I**

may be a varistor, for example. However, the voltage clamping element **2I** does not have to be a varistor but may also be a Zener diode (such as a TVS diode).

(136) Next, an exemplary operation of the switch system **20I** will be described with reference to FIG. **14**. In FIG. **14**, the waveform labeled as “semiconductor switch (**Tr1**)” indicates how the first control signal supplied to the control terminal **10** of the semiconductor switch **1** (junction field effect transistor **Tr1**) changes with time. Also, in FIG. **14**, the waveform labeled as “control switch (**Q1**)” indicates how the state of the control switch **33** (field effect transistor **Q1**) changes with time. Furthermore, in FIG. **14**, “**Vgs**” denotes a voltage between the gate terminal **G1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor **Tr1**. Thus, “**Vgs**” denotes the voltage between the control terminal **10** and second main terminal **12** of the semiconductor switch **1**. Furthermore, in FIG. **14**, “**Vds**” denotes a voltage between the drain terminal **D1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor **Tr**. Thus, “**Vds**” denotes the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. The voltage clamping element **2** is connected to the semiconductor switch **1** in parallel. Thus, the voltage across the voltage clamping element **2** is the same as the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(137) FIG. **14** illustrates how the switch system **20I** operates, for example, in a situation where a short-circuit failure occurs to the load circuit **204** at a time **t31** to cause the main current **I1** of the semiconductor switch **1** to start to increase.

(138) At a time **t32**, the first control signal output from the first control unit **4** changes from a first on signal into a first off signal. In response, the semiconductor switch **1** is turned off and the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** increases. Then, the gate of the field effect transistor **Q1** is charged through the capacitor **55** of the second control unit **5I** to turn the field effect transistor **Q1** on. In the switch system **20I**, when the field effect transistor **Q1** turns on, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** stops increasing, thus turning the field effect transistor **Q1** off. In the switch system **20I**, when the field effect transistor **Q1** turns off, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** increases, and the gate of the field effect transistor **Q1** is charged through the capacitor **55** of the second control unit **5I**, thus turning the field effect transistor **Q1** on. In the switch system **20I**, the field effect transistor **Q1** alternately turns on and off repeatedly, thus causing an increase in the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. In the switch system **20I**, when the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** increases to reach the clamp voltage of the voltage clamping element **2I** at a time **t33**, the supply of power to the gate of the field effect transistor **Q1** via the capacitor **55** is stopped. That is to say, in the switch system **20I**, the active clamping circuit **3I** finishes performing the clamping operation. As a result, in the switch system **20I**, the voltage clamping element **2I** clamps the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. In the switch system **20I**, in the period between the times **t32** and **t33**, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. The main current **I1** flowing through the semiconductor switch **1** starts decreasing at the time **t33**.

(139) At the time **t33**, the voltage clamping element **2I** clamps the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. Then, the absolute value of a current variation ratio ($-dI1/dt$) of the main current **I1** of the semiconductor switch **1** increases to cause the main current **I1** to decrease steeply to zero. When the main current **I1** flowing through the semiconductor switch **1** goes zero, the current **I2** flowing through the voltage clamping element **2I** decreases. Thereafter, at a time **t34**, the current **I2** flowing through the voltage clamping element **2I** goes zero and the voltage **Vds** between the first main terminal **11** and second

main terminal **12** of the semiconductor switch **1** becomes a constant voltage lower than the clamp voltage.

(140) The switch system **20I** may make the current value of the main current **I1** flowing through the semiconductor switch **1** at the time **t33** smaller than the current value of the main current **I1** flowing through the semiconductor switch **1** at the time **t32**. This allows the switch system **20I** to reduce the clamp voltage of the voltage clamping element **2I** and thereby reduce the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(141) A switch system **20I** according to the tenth embodiment described above includes a semiconductor switch **1**, a voltage clamping element **2I**, an active clamping circuit **3I**, a first control unit **4**, and a second control unit **5I**. The semiconductor switch **1** has a control terminal **10**, a first main terminal **11**, and a second main terminal **12**. The voltage clamping element **2I** is connected to the semiconductor switch **1** in parallel. The active clamping circuit **3I** is connected between the control terminal **10** and the first main terminal **11** of the semiconductor switch **1**. The first control unit **4** controls the semiconductor switch **1**. The second control unit **5I** controls the active clamping circuit **3I**. The active clamping circuit **3I** includes a first diode **31**, a second diode **32**, and a control switch **33**. The first diode **31** has a first anode and a first cathode. The first diode **31** causes breakdown upon application of voltage between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1**. The second diode **32** has a second anode and a second cathode. The second cathode of the second diode **32** is connected to the first cathode of the first diode **31**. The control switch **33** is connected between the first anode of the first diode **31** and the control terminal **10** of the semiconductor switch **1**. The control switch **33** is a field effect transistor **Q1** having a gate, a drain, and a source. The drain of the field effect transistor **Q1** is connected to the first anode of the first diode **31**. The source of the field effect transistor **Q1** is connected to the control terminal **10** of the semiconductor switch **1**. The second control unit **5I** includes a capacitor **55** and a Zener diode **56**. The capacitor **55** has a first terminal and a second terminal. The first terminal of the capacitor **55** is connected to the gate of the field effect transistor **Q1**. The second terminal of the capacitor **55** is connected to the first cathode of the first diode **31** and the second cathode of the second diode **32**. The Zener diode **56** has a third anode and a third cathode. The third cathode of the Zener diode **56** is connected to the first terminal of the capacitor **55** and the gate of the field effect transistor **Q1**. The third anode of the Zener diode **56** is connected to the source of the field effect transistor **Q1**.

(142) The switch system **20I** according to the tenth embodiment may reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off. More specifically, the switch system **20I** according to the tenth embodiment may reduce the current value of the main current **I1** flowing through the semiconductor switch **1** when the voltage clamping element **2I** clamps the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** (i.e., may reduce the clamp voltage). Consequently, the switch system **20I** may reduce deterioration in the characteristics of the semiconductor switch **1** (e.g., may reduce the chances of the semiconductor switch **1** coming to have a shorter life).

Eleventh Embodiment

(143) Next, a switch system **20J** according to an eleventh embodiment will be described with reference to FIGS. **15** and **16**.

(144) The switch system **20J** according to the eleventh embodiment is substantially the same as the switch system **20F** according to the seventh embodiment (refer to FIG. **8**) but further includes a detection circuit **14** and a switch element **15**, which is a difference from the switch system **20F** according to the seventh embodiment. In addition, the switch system **20J** according to the eleventh embodiment includes the NOT circuit **7G** of the switch system **20G** according to the eighth embodiment instead of the NOT circuit **7F** of the switch system **20F** according to the seventh embodiment (refer to FIG. **8**), which is another difference from the switch system **20F** according to the seventh embodiment. In the following description, any constituent element of the switch system

20J according to this eleventh embodiment, having the same function as a counterpart of the switch system **20F** according to the seventh embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(145) The detection circuit **14** is connected between the control terminal **10** and the second main terminal **12** of the semiconductor switch **1**. The detection circuit **14** detects voltage between the control terminal **10** and the second main terminal **12** of the semiconductor switch **1**. The detection circuit **14** includes a Zener diode **142** and a resistor **141**. The Zener diode **142** has an anode and a cathode. The cathode of the Zener diode **142** is connected to the control terminal **10** of the semiconductor switch **1** and the anode of the Zener diode **142** is connected to the second main terminal **12** of the semiconductor switch **1** via the resistor **141**.

(146) The switch element **15** is connected between the control terminal **10** and the second main terminal **12** of the semiconductor switch **1**. The switch element **15** is a field effect transistor **Q5** having a gate, a drain, and a source. In the field effect transistor **Q5**, the drain thereof is connected to the control terminal **10** of the semiconductor switch **1**, the source thereof is connected to the second main terminal **12** of the semiconductor switch **1**, and the gate thereof is connected to the anode of the Zener diode **142**. Thus, the resistor **141** is connected between the gate and source of the field effect transistor **Q5**.

(147) In the switch system **20J**, the switch element **15** turns on when the voltage detected by the detection circuit **14** exceeds a threshold voltage V_{th} . The threshold voltage V_{th} of the detection circuit **14** may be determined by a Zener voltage of the Zener diode **142**. The threshold voltage V_{th} may be 5 V, for example, which is larger than the gate voltage V_{g1} (of 3 V, for example) of the junction field effect transistor **Tr** in on state and smaller than the output voltage (of 15 V, for example) of the first DC power supply included in the first control unit **4**.

(148) In the switch system **20J**, the semiconductor switch **1** is a junction field effect transistor **Tr1**. If a MOSFET were adopted as the semiconductor switch **1**, then the output voltage of the first DC power supply of the first control unit **4** would be equal to the gate voltage of the MOSFET in on state. The threshold voltage V_{th} of the detection circuit **14** should be set at a value (e.g., 20V) larger than the output voltage of the first DC power supply included in the first control unit **4** in that case.

(149) In the switch system **20J** on the other hand, the semiconductor switch **1** is a junction field effect transistor **Tr1**. Thus, if neither the load circuit **204** nor the junction field effect transistor **Tr** has caused any error, then the gate voltage V_{g1} of the junction field effect transistor **Tr** in on state is smaller than the output voltage of the first DC power supply of the first control unit **4**, as for the gate voltage V_{gs} of the junction field effect transistor **Tr1**. The voltage value of the first on signal of the first control signal output from the first control unit **4** is the same as the output voltage of the first DC power supply of the first control unit **4**. In the switch system **20J**, the semiconductor switch **1** is the junction field effect transistor **Tr1**. Thus, the threshold voltage V_{th} for use in the detection circuit **14** may be a voltage value larger than the gate voltage V_{g1} of the junction field effect transistor **Tr** in on state and smaller than the output voltage of the first DC power supply of the first control unit **4**. The threshold voltage V_{th} does not have to be 5 V but may also be a voltage value falling within the range from 6 V to 14 V, for example.

(150) Next, an exemplary operation of the switch system **20J** will be described with reference to FIG. **16**. In FIG. **16**, the waveform labeled as "semiconductor switch (**Tr1**)" indicates how the first control signal supplied to the control terminal **10** of the semiconductor switch **1** (junction field effect transistor **Tr1**) changes with time. Also, in FIG. **16**, the waveform labeled as "control switch (**Q1**)" indicates how the second control signal supplied to the control switch **33** (field effect transistor **Q1**) changes with time. In FIG. **16**, the waveform labeled as "second FET (**Q2**)" indicates how a third control signal supplied to the second gate of the second field effect transistor **Q2** changes with time. In FIG. **16**, the waveform labeled as "third FET (**Q3**)" indicates how a control signal supplied to the third gate of the third field effect transistor **Q3** changes with time. In FIG. **16**,

the waveform labeled as “switch element (Q5)” indicates how a control signal supplied to the gate of the field effect transistor Q5 changes with time. Furthermore, in FIG. 16, “Vgs” denotes a voltage between the gate terminal G1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vgs” denotes the voltage between the control terminal 10 and second main terminal 12 of the semiconductor switch 1 and denotes the gate voltage of the junction field effect transistor Tr1. Furthermore, in FIG. 16, “Vds” denotes a voltage between the drain terminal D1 and the source terminal S1 with respect to the source terminal S1 of the junction field effect transistor Tr1. Thus, “Vds” denotes the voltage between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1. The voltage clamping element 2 is connected to the semiconductor switch 1 in parallel. Thus, the voltage across the voltage clamping element 2 is the same as the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1.

(151) FIG. 16 illustrates how the switch system 20J operates, for example, in a situation where a short-circuit failure occurs to the load circuit 204 at a time t41 to cause the main current I1 of the semiconductor switch 1 to start to increase.

(152) At a time t42, the first control signal changes from a first on signal into a first off signal, the second control signal changes from a second off signal into a second on signal, and the control signal for the third field effect transistor Q3 changes from an on signal into an off signal. This causes not only the semiconductor switch 1 to start turning off but also the control switch 33 (first field effect transistor Q1) of the active clamping circuit 3A to turn on. When the control switch 33 of the active clamping circuit 3A turns on, the voltage Vds between the first main terminal 11 and second main terminal 12 of the semiconductor switch 1 is clamped by the active clamping circuit 3A. As a result, part of the energy stored in the inductance component of the line 203 of the load circuit 204 is consumed by the semiconductor switch 1. The main current I1 flowing through the semiconductor switch 1 starts to decrease at the time t42.

(153) Before a time t44, when the predetermined period TO passes since the time t42, the gate voltage Vgs starts to increase. Thereafter, when the gate voltage Vgs detected by the detection circuit 14 becomes equal to or larger than the threshold voltage Vth at a time t43, an on signal is supplied from the detection circuit 14 to the switch element 15 to turn the switch element Q5 on. In the switch system 20J, when the switch element 15 turns on, the gate voltage Vgs of the junction field effect transistor Tr goes zero, the semiconductor switch 1 turns off, and the main current I1 of the semiconductor switch 1 goes zero. In the period from the time t43 to the time t44, the main current I1 that flowed through the semiconductor switch 1 may either become a current flowing through only the first diode 31, out of the first diode 31 and the voltage clamping element 2 (hereinafter referred to as a “first case”) or be shunted into two currents flowing through the first diode 31 and the voltage clamping element 2 separately (hereinafter referred to as a “second case”). In FIG. 16, the first case is illustrated. The first case is a situation where the first diode 31 may maintain an amount of current flowing through the semiconductor switch 1. If the switch system 20J needs to operate in the first case scenario, components that allow the amount of current expected to flow through the semiconductor switch 1 may be selected as for the first diode 31 and the control switch 33. As the first diode 31, a varistor may be used, for example. In the second case, the first diode 31 is a diode, of which the specification prevents the first diode 31 from allowing the amount of current that flowed through the semiconductor switch 1 to flow therethrough. Thus, in the second case, the current that flowed through the semiconductor switch 1 flows through the voltage clamping element 2 and the clamp voltage of the voltage clamping element 2 is applied to the semiconductor switch 1. Also, the clamp voltage of the voltage clamping element 2 is also applied to a series circuit of the first diode 31, the second diode 32, the control switch 33, and the switch element 15. In the series circuit, the voltage applied to the first diode 31 is the greatest. Therefore, a diode, of which the peak inverse voltage is larger than the clamp voltage of the voltage clamping element 2, is preferably selected as the first diode 31. In addition, in each of the first and

second cases, in the period from the time **t43** to the time **t44**, a large amount of current, which is significantly different from the current flowing when the active clamping circuit **3A** is in a normal operating state as in the period from the time **t42** to the time **t43**, flows through the active clamping circuit **3A**. Alternatively, in the period from the time **t43** to the time **t44**, a high voltage, which is significantly different from the voltage applied when the active clamping circuit **3A** is in a normal operating state as in the period from the time **t42** to the time **t43**, is applied to the active clamping circuit **3A**. That is why the period from the time **t43** to the time **t44** is preferably as short as possible. Therefore, when the switch system **20J** is designed, the predetermined period **TO** may be determined to reduce deterioration in the characteristics of the active clamping circuit **3A** in the period from the time **t43** to the time **t44**, for example, and the circuit constant of the second control unit **5E** may be set based on the predetermined period **TO**. In the switch system **20J**, at the time **t44**, the on signal is input to the second field effect transistor **Q2** to turn the second field effect transistor **Q2** on, thus causing the control switch **33** to turn off. Thereafter, when the current **I2** of the voltage clamping element **2** goes zero at the time **t45**, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** becomes a constant voltage lower than the clamp voltage.

(154) In the switch system **20J**, as well as in the switch system **20F** according to the seventh embodiment (refer to FIG. **8**), the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, in the switch system **20J**, when the control switch **33** (first field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3A**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20J**, when the switch element **15** turns on before the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2**. Also, in the switch system **20J**, unless the switch element **15** turns on before the control switch **33** of the active clamping circuit **3A** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2** when the control switch **33** turns from on to off.

(155) The switch system **20J** according to the eleventh embodiment, as well as the switch system **20F** according to the seventh embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off. In addition, the switch system **20J** according to the eleventh embodiment may also reduce deterioration in the characteristics of the semiconductor switch **1** due to an increase in the voltage (gate voltage **Vgs**) between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1** while the active clamping circuit **3A** is performing the clamping operation.

Twelfth Embodiment

(156) Next, a switch system **20K** according to a twelfth embodiment will be described with reference to FIG. **17**.

(157) The switch system **20K** according to the twelfth embodiment is substantially the same as the switch system **20G** according to the eighth embodiment (refer to FIG. **10**) but includes a semiconductor switch **1K** instead of the semiconductor switch **1** of the switch system **20G**, which is a difference from the switch system **20G**. In addition, the switch system **20K** according to the twelfth embodiment includes a plurality of (e.g., two) gate resistors **R20**, which is another difference from the switch system **20G** according to the eighth embodiment. In the following description, any constituent element of the switch system **20K** according to this twelfth embodiment, having the same function as a counterpart of the switch system **20G** according to the eighth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(158) The semiconductor switch **1K** is made up of a plurality of (e.g., two) field effect transistors **Tr2** which are connected in parallel. Each of the plurality of field effect transistors **Tr2** has a gate, a drain, and a source. In the semiconductor switch **1K**, the first main terminal **11** is connected to the respective drains of the plurality of field effect transistors **Tr2** and the second main terminal **12** is connected to the respective sources of the plurality of field effect transistors **Tr2**.

(159) The plurality of gate resistors **R20** correspond one to one to the respective gates of the semiconductor switch **1K** and are connected between the first control unit **4** and the respective gates. The plurality of gate resistors **R20** are connected to the active clamping circuit **3A**.

(160) The switch system **20K** according to the twelfth embodiment, as well as the switch system **20G** according to the eighth embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

(161) In the switch system **20K** according to the twelfth embodiment, the semiconductor switch **1K** is made up of a plurality of field effect transistors **Tr2** which are connected in parallel. This may make the current value of the current flowing through each of the plurality of field effect transistors **Tr2** smaller than the current value of the current flowing through the junction field effect transistor **Tr1**.

(162) In addition, the switch system **20K** according to the twelfth embodiment includes the plurality of gate resistors **R20**. This enables increasing the impedance of the loop, including the drain and gate, of each of the plurality of field effect transistors **Tr2** compared to a situation where the plurality of gate resistors **R20** are not provided, thus reducing the chances of producing oscillation.

Thirteenth Embodiment

(163) Next, a switch system **20L** according to a thirteenth embodiment will be described with reference to FIG. **18**.

(164) The switch system **20L** according to the thirteenth embodiment is substantially the same as the switch system **20G** according to the eighth embodiment (refer to FIG. **10**) but includes a semiconductor switch **1L** instead of the semiconductor switch **1** of the switch system **20G**, which is a difference from the switch system **20G**. The semiconductor switch **1L** is a bidirectional switch. In addition, the switch system **20L** according to the thirteenth embodiment includes the voltage clamping element **2I** instead of the voltage clamping element **2**, which is another difference from the switch system **20G**. Furthermore, the switch system **20L** includes two active clamping circuits **3A**, two first control units **4**, two NOT circuits **7**, two second control units **5E**, and two first drive circuits **8**, which is still another difference from the switch system **20G** according to the eighth embodiment. Besides, the switch system **20L** according to the thirteenth embodiment does not include the third drive circuit **13G** of the switch system **20G** according to the eighth embodiment, which is yet another difference from the switch system **20G** according to the eighth embodiment. In the following description, any constituent element of the switch system **20L** according to this thirteenth embodiment, having the same function as a counterpart of the switch system **20G** according to the eighth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(165) In the switch system **20L**, the power supply **202** of the load circuit **204** connected between the first external terminal **Tm1** and the second external terminal **Tm2** may be an AC power supply, for example.

(166) In the switch system **20L**, the semiconductor switch **1L** includes a first switch element **101** and a second switch element **102**. Each of the first switch element **101** and the second switch element **102** may be a junction field effect transistor, for example. The junction field effect transistor may be a GaN-based GIT, for example.

(167) The first switch element **101** includes a first gate terminal **G11**, a first drain terminal **D11**, and a first source terminal **S11**.

(168) The second switch element **102** includes a second gate terminal **G12**, a second drain terminal

D12, and a second source terminal **S12**.

(169) The second switch element **102** is connected to the first switch element **101** in series. In the semiconductor switch **1L**, the first drain terminal **D11** of the first switch element **101** and the second drain terminal **D12** of the second switch element **102** are connected to each other.

(170) The semiconductor switch **1L** includes two sets, each of which consists of a control terminal, a first main terminal, and a second main terminal. In one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal **G11**, the second source terminal **S12**, and the first source terminal **S11**, respectively. In the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal **G12**, the first source terminal **S11**, and the second source terminal **S12**, respectively.

(171) The voltage clamping element **2I** is a varistor. The voltage clamping element **2I** is connected to the semiconductor switch **1L** in parallel. More specifically, the voltage clamping element **2I** is connected between the first source terminal **S11** and the second source terminal **S12** of the semiconductor switch **1L**.

(172) In the switch system **20L**, one first control unit **4** out of the two first control units **4** is connected between the first gate terminal **G11** and the first source terminal **S11** of the first switch element **101**, while the other first control unit **4** is connected between the second gate terminal **G12** and the second source terminal **S12** of the second switch element **102**.

(173) The switch system **20L** includes two active clamping circuits **3A** as described above. In the switch system **20L**, one active clamping circuit **3A** out of the two active clamping circuits **3A** is connected between the first gate terminal **G11** and the second source terminal **S12**, while the other active clamping circuit **3A** is connected between the second gate terminal **G12** and the first source terminal **S11**.

(174) The switch system **20L** according to the thirteenth embodiment may reduce, while a main current is flowing from the first source terminal **S11** of the semiconductor switch **1L** toward the second source terminal **S12** thereof, the surge voltage applied to the semiconductor switch **1L** when the semiconductor switch **1L** turns off. In addition, the switch system **20L** according to the thirteenth embodiment may also reduce, while a main current is flowing from the second source terminal **S12** of the semiconductor switch **1L** toward the first source terminal **S11** thereof, the surge voltage applied to the semiconductor switch **1L** when the semiconductor switch **1L** turns off.

Fourteenth Embodiment

(175) Next, a switch system **20M** according to a fourteenth embodiment will be described with reference to FIG. **19**.

(176) The switch system **20M** according to the fourteenth embodiment is substantially the same as the switch system **20G** according to the eighth embodiment (refer to FIG. **10**) but includes two semiconductor switches **1** and two voltage clamping elements **2** of the switch system **20G**, which is a difference from the switch system **20G** according to the eighth embodiment. In addition, the switch system **20M** according to the fourteenth embodiment includes two active clamping circuits **3A**, which is another difference from the switch system **20G** according to the eighth embodiment. Furthermore, the switch system **20M** according to the fourteenth embodiment does not include the third drive circuit **13G** of the switch system **20G** according to the eighth embodiment, which is yet another difference from the switch system **20G** according to the eighth embodiment. Besides, the switch system **20M** according to the fourteenth includes a NOT circuit **7M** instead of the NOT circuit **7G** of the switch system **20G**, which is yet another difference from the switch system **20G** according to the eighth embodiment. In the following description, any constituent element of the switch system **20M** according to this fourteenth embodiment, having the same function as a counterpart of the switch system **20G** according to the eighth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(177) In each of the two semiconductor switches **1**, the control terminal **10**, the first main terminal **11**, and the second main terminal **12** are a gate terminal **G1**, a drain terminal **D1**, and a source

terminal **S1**, respectively. In this switch system **20M**, the two semiconductor switches **1** are connected together in series by connecting their respective source terminals **S1** to each other. Thus, the switch system **20M** includes a bidirectional switch including the two semiconductor switches **1**. In the switch system **20M**, the power supply **202** of the load circuit **204** connected between the first external terminal **Tm1** and the second external terminal **Tm2** may be an AC power supply, for example.

(178) The two voltage clamping elements **2** correspond one to one to the two semiconductor switches **1**. Each of the two voltage clamping elements **2** is connected to a corresponding one of the two semiconductor switches **1** in parallel. The two active clamping circuits **3A** correspond one to one to the two semiconductor switches **1**. Each of the two active clamping circuits **3A** is connected between the gate terminal **G1** and drain terminal **D1** of the corresponding one of the two semiconductor switches **1**. Also, in this switch system **20M**, the second field effect transistor **Q2** is shared by the two active clamping circuits **3A**.

(179) The NOT circuit **7M** includes a diode **73** instead of the resistor **72** of the NOT circuit **7G** of the switch system **20G**. The diode **73** may be formed by, for example, connecting together the gate and source of a normally-on junction field effect transistor.

(180) The switch system **20M** according to the fourteenth embodiment may reduce, while a main current is flowing through each semiconductor switch **1** from the first external terminal **Tm1** toward the second external terminal **Tm2**, the surge voltage applied to each semiconductor switch **1** when the semiconductor switch **1** turns off. In addition, the switch system **20M** according to the fourteenth embodiment may also reduce, while a main current is flowing through each semiconductor switch **1** from the second external terminal **Tm2** toward the first external terminal **Tm1**, the surge voltage applied to each semiconductor switch **1** when the semiconductor switch **1L** turns off.

Fifteenth Embodiment

(181) Next, a switch system **20N** according to a fifteenth embodiment will be described with reference to FIG. **20**.

(182) The switch system **20N** according to the fifteenth embodiment is substantially the same as the switch system **20L** according to the thirteenth embodiment (refer to FIG. **18**) but includes a semiconductor switch **1N** instead of the semiconductor switch **1L** of the switch system **20L**, which is a difference from the switch system **20L**. The semiconductor switch **1N** is a dual-gate bidirectional switch. In the following description, any constituent element of the switch system **20N** according to this fifteenth embodiment, having the same function as a counterpart of the switch system **20L** according to the thirteenth embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(183) In the switch system **20N**, the power supply **202** of the load circuit **204** connected between the first external terminal **Tm1** and the second external terminal **Tm2** may be an AC power supply, for example.

(184) The semiconductor switch **1N** is a dual-gate bidirectional switch having a first gate terminal **G11**, a first source terminal **S11** corresponding to the first gate terminal **G11**, a second gate terminal **G12**, and a second source terminal **S12** corresponding to the second gate terminal **G12**. The semiconductor switch **1N** includes two sets, each of consists of a control terminal, a first main terminal, and a second main terminal. In one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal **G11**, the second source terminal **S12**, and the first source terminal **S11**, respectively. In the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal **G12**, the first source terminal **S11**, and the second source terminal **S12**, respectively.

(185) The semiconductor switch **1N** is a type of GaN-based GIT. The semiconductor switch **1N** includes, for example, a substrate, a buffer layer, a first nitride semiconductor layer, a second nitride semiconductor layer, a first source electrode, a first gate electrode, a second gate electrode, a

second source electrode, a first p-type layer, and a second p-type layer. The buffer layer is formed on the substrate. The first nitride semiconductor layer is formed on the buffer layer. The second nitride semiconductor layer is formed on the first nitride semiconductor layer. The first source electrode, the first gate electrode, the second gate electrode, and the second source electrode are formed on the second nitride semiconductor layer. The first p-type layer is interposed between the first gate electrode and the second nitride semiconductor layer. The second p-type layer is interposed between the second gate electrode and the second nitride semiconductor layer. In the semiconductor switch **1N**, the first source terminal **S11** includes the first source electrode. The first gate terminal **G11** includes the first gate electrode and the first p-type layer. The second gate terminal **G12** includes the second gate electrode and the second p-type layer. The second source terminal **S12** includes the second source electrode. The substrate may be a silicon substrate, for example. The buffer layer may be an undoped GaN layer, for example. The first nitride semiconductor layer may be, for example, an undoped GaN layer. The second nitride semiconductor layer may be, for example, an undoped AlGaN layer. Each of the first and second p-type layers may be, for example, a p-type AlGaN layer. Each of the buffer layer, the first nitride semiconductor layer, and the second nitride semiconductor layer may include impurities such as Mg, H, Si, C, and O to be inevitably contained during their growing process by metal-organic vapor phase epitaxy (MOVPE), for example.

(186) In the semiconductor switch **TN**, the second nitride semiconductor layer forms a heterojunction portion along with the first nitride semiconductor layer. In the first nitride semiconductor layer, a two-dimensional electron gas has been generated in the vicinity of the heterojunction portion. A region including the two-dimensional electron gas (hereinafter referred to as a “two-dimensional electron gas layer”) may also serve as an n-channel layer (electron conduction layer).

(187) Also, in the following description, a state where a voltage equal to or higher than a first threshold voltage (of 1.3 V, for example) is not applied between the first gate terminal **G11** and the first source terminal **S11** will be hereinafter referred to as a “state where the first gate terminal **G11** is off.” Also, a state where a voltage equal to or higher than the first threshold voltage is applied between the first gate terminal **G11** and the first source terminal **S11** with the first gate terminal **G11** having the higher potential will be hereinafter referred to as a “state where the first gate terminal **G11** is on.” Furthermore, a state where a voltage equal to or higher than a second threshold voltage (of 1.3 V, for example) is not applied between the second gate terminal **G12** and the second source terminal **S12** will be hereinafter referred to as a “state where the second gate terminal **G12** is off.” Also, a state where a voltage equal to or higher than the second threshold voltage is applied between the second gate terminal **G12** and the second source terminal **S12** with the second gate terminal **G12** having the higher potential will be hereinafter referred to as a “state where the second gate terminal **G12** is on.”

(188) This semiconductor switch **1N** includes the first p-type layer and the second p-type layer, thus implementing a normally off transistor.

(189) The semiconductor switch **1N** may be switched from one of a bidirectionally on state, a bidirectionally off state, a first diode state, or a second diode state to another depending on the combination of a first gate voltage applied to the first gate terminal **G11** and a second gate voltage applied to the second gate terminal **G12**. The first gate voltage is a voltage applied between the first gate terminal **G11** and the first source terminal **S11**. The second gate voltage is a voltage applied between the second gate terminal **G12** and the second source terminal **S12**. The bidirectionally on state is a state where an electric current is allowed to pass bidirectionally (i.e., in a first direction and a second direction opposite from the first direction). The bidirectionally off state is a state where an electric current is blocked bidirectionally. The first diode state is a state where an electric current is allowed to pass in the first direction. The second diode state is a state where an electric current is allowed to pass in the second direction.

(190) In a state where the first gate terminal **G11** is on and the second gate terminal **G12** is on, the semiconductor switch **TN** turns into the bidirectionally on state. In a state where the first gate terminal **G11** is off and the second gate terminal **G12** is off, the semiconductor switch **TN** turns into the bidirectionally off state. In a state where the first gate terminal **G11** is off and the second gate terminal **G12** is on, the semiconductor switch **TN** turns into the first diode state. In a state where the first gate terminal **G11** is on and the second gate terminal **G12** is off, the semiconductor switch **TN** turns into the second diode state.

(191) The switch system **20N** includes two active clamping circuits **3A**. In the switch system **20N**, one active clamping circuit **3A** out of the two active clamping circuits **3A** is connected between the first gate terminal **G11** and the second source terminal **S12**, while the other active clamping circuit **3A** is connected between the second gate terminal **G12** and the first source terminal **S11**.

(192) The switch system **20N** according to the fifteenth embodiment may reduce, while a main current is flowing from the first source terminal **S11** of the semiconductor switch **TN** toward the second source terminal **S12** thereof, the surge voltage applied to the semiconductor switch **TN** when the semiconductor switch **TN** turns off. In addition, the switch system **20N** according to the fifteenth embodiment may also reduce, while a main current is flowing from the second source terminal **S12** of the semiconductor switch **TN** toward the first source terminal **S1** thereof, the surge voltage applied to the semiconductor switch **TN** when the semiconductor switch **TN** turns off.

Sixteenth Embodiment

(193) Next, a switch system **20O** according to a sixteenth embodiment will be described with reference to FIG. **21**.

(194) The switch system **20O** according to the sixteenth embodiment is substantially the same as the switch system **20** according to the first embodiment (refer to FIG. **1**) but further includes a current detection unit **16**, which is a difference from the switch system **20**. In addition, the switch system **20O** according to the sixteenth embodiment includes an active clamping circuit **30** instead of the active clamping circuit **3** of the switch system **20** according to the first embodiment, which is another difference from the switch system **20**. Furthermore, the switch system **20O** according to the sixteenth embodiment includes the voltage clamping element **2I** instead of the voltage clamping element **2**, which is another difference from the switch system **20**. In the following description, any constituent element of the switch system **20O** according to this sixteenth embodiment, having the same function as a counterpart of the switch system **20** according to the first embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(195) The current detection unit **16** detects a current flowing through the second main terminal **12** of the semiconductor switch **1**. The current detection unit **16** is connected between the second main terminal **12** of the semiconductor switch **1** and the second external terminal **Tm2**.

(196) The active clamping circuit **30** is connected between the control terminal **10** and the first main terminal **11** of the semiconductor switch **1**. The active clamping circuit **30** includes a first diode **31**, a second diode **32**, and a control switch **33**. In the active clamping circuit **30**, the second anode of the second diode **32** is connected to the first main terminal **11** of the semiconductor switch **1**, and the second cathode of the second diode **32** is connected to the control switch **33** via the first diode **31**. The second cathode of the second diode **32** is connected to the first cathode of the first diode **31**. The control switch **33** is connected between the first anode of the first diode **31** and the control terminal **10** of the semiconductor switch **1**. The first diode **31** causes breakdown upon application of voltage between the first main terminal **11** and the second main terminal **12** of the semiconductor switch **1**. The control switch **33** is a field effect transistor **Q1**. The drain of the field effect transistor **Q1** is connected to the first anode of the first diode **31**. The source of the field effect transistor **Q1** is connected to the control terminal **10** of the semiconductor switch **1**.

(197) The voltage clamping element **2I** is connected to the semiconductor switch **1** in parallel. More specifically, the voltage clamping element **2I** is connected between the first main terminal **11**

and the second main terminal **12** of the semiconductor switch **1**. The voltage clamping element **2I** may be a varistor, for example. However, the voltage clamping element **2I** does not have to be a varistor but may also be a Zener diode (such as a TVS diode).

(198) In the switch system **20O**, the second control unit **5** controls the control switch **33** according to the output of the current detection unit **16**. More specifically, the second control unit **5** changes the conductive period of the control switch **33** according to the output of the current detection unit **16**. When finding the current value detected by the current detection unit **16** while the active clamping circuit **30** is performing the clamping operation larger than a predetermined value, the larger the current value is, the shorter the conductive period of the control switch **33** is made by the second control unit **5**. When the current value detected by the current detection unit **16** becomes equal to or larger than a threshold value, the second control unit **5** turns the control switch **33** on to make the active clamping circuit **30** start performing the clamping operation. When finding that the current value detected by the current detection unit **16** while the active clamping circuit **30** is performing the clamping operation reaches a predetermined value larger than the threshold value, the larger the current value is, the earlier the second control unit **5** performs the control of turning the control switch **33** from on to off. As used herein, the predetermined value is a current value falling within a safe operating range of the voltage clamping element **2I**.

(199) In the switch system **20O**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2I**. Thus, in the switch system **20O**, when the control switch **33** (field effect transistor **Q1**) is turned on to turn the semiconductor switch **1** off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **30**. As a result, part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**. In the switch system **20O**, when the control switch **33** of the active clamping circuit **30** turns from on to off, the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the voltage clamping element **2I**.

(200) The switch system **20O** according to the sixteenth embodiment, as well as the switch system **20** according to the first embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Seventeenth Embodiment

(201) Next, a switch system **20P** according to a seventeenth embodiment will be described with reference to FIG. **22**.

(202) The switch system **20P** according to the seventeenth embodiment is substantially the same as the switch system **20F** according to the seventh embodiment (refer to FIG. **8**) but further includes a current detection unit **16P**, which is a difference from the switch system **20F**. In addition, the switch system **20P** includes a second control unit **5P** instead of the second control unit **5E** of the switch system **20F**, which is another difference from the switch system **20F**. Furthermore, the switch system **20P** includes the NOT circuit **7G** of the switch system **20G** according to the eighth embodiment (refer to FIG. **10**) instead of the NOT circuit **7F** of the switch system **20F**, which is still another difference from the switch system **20F**. In the following description, any constituent element of the switch system **20P** according to this seventeenth embodiment, having the same function as a counterpart of the switch system **20F** according to the seventh embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(203) The current detection unit **16P** detects a current flowing through the second main terminal **12** of the semiconductor switch **1**.

(204) The second control unit **5P** includes a third field effect transistor **Q13**. The third field effect transistor **Q13** has a third gate, a third drain, and a third source.

(205) The current detection unit **16P** includes a detecting resistor **161** and an integration circuit **162**. The detecting resistor **161** is connected to the second main terminal **12** of the semiconductor

switch **1** and connected to the semiconductor switch **1** in series. More specifically, the detecting resistor **161** is connected between the second main terminal **12** of the semiconductor switch **1** and the second external terminal Tm2. The integration circuit **162** integrates voltage across the detecting resistor **161**. The integration circuit **162** includes an operational amplifier **163** and a capacitor **165**. The integration circuit **162** further includes a biasing resistor **164**. In the integration circuit **162**, the second main terminal **12** of the semiconductor switch **1** is connected to a non-inverting input terminal of the operational amplifier **163**, and the capacitor **165** is connected between an inverting input terminal and output terminal operational amplifier **163**. The output terminal of the operational amplifier **163** is connected to the second gate of the second field effect transistor Q2. Also, in the integration circuit **162**, the biasing resistor **164** is connected between the inverting input terminal of the operational amplifier **163** and the second external terminal Tm2.

(206) In the switch system **20P**, the second gate of the second field effect transistor Q2 is connected to the output terminal of the integration circuit **162**. In the switch system **20P**, the third drain of the third field effect transistor Q13 is connected to the output terminal of the integration circuit **162**. In the switch system **20P**, the third gate of the third field effect transistor Q13 is connected to a signal path between the first control unit **4** and the control terminal **10** of the semiconductor switch **1**.

(207) In the switch system **20P**, while the first control unit **4** is outputting a first on signal, the semiconductor switch **1** and the third field effect transistor Q13 are in on state, and therefore, the output terminal and inverting input terminal of the operational amplifier **163** are short-circuited. When the first control signal output from the first control unit **4** changes from the first on signal into a first off signal, the third field effect transistor Q13 turns off. In the current detection unit **16P**, the integration circuit **162** starts performing the integration operation, the instant the third field effect transistor Q13 turns off. In the switch system **20P**, when the integrated value of the integration circuit **162** reaches a threshold value, the output signal of the integration circuit **162** causes the second field effect transistor Q2 to turn on. In the active clamping circuit **3A**, when the second field effect transistor Q2 turns on, the first field effect transistor Q1 turns off, thus ending the clamping operation.

(208) The current detection unit **16** includes the integration circuit **162**. Thus, the time it takes for the potential at the non-inverting input terminal of the operational amplifier **163** to become higher than the potential at the inverting input terminal (e.g., 6V) varies according to the current value of the current flowing through the detecting resistor **161**. The biasing resistor **164** is provided to bias the potential at the non-inverting input terminal of the operational amplifier **163** (i.e., a reference potential) at a predetermined potential (e.g., 6 V). In the current detection unit **16P**, the larger the current value of the current flowing through the detecting resistor **161** is, the shorter the time it takes to turn the second field effect transistor Q2 on is and the shorter the time it takes to turn the first field effect transistor Q1 off. This allows the switch system **20P** to change the conductive period of the control switch **33** of the active clamping circuit **3A** according to the current value of the current flowing through the detecting resistor **161**.

(209) The switch system **20P** according to the seventeenth embodiment, as well as the switch system **20F** according to the seventh embodiment, may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Eighteenth Embodiment

(210) Next, a switch system **20Q** according to an eighteenth embodiment will be described with reference to FIGS. **23** and **24**.

(211) The switch system **20Q** according to the eighteenth embodiment is substantially the same as the switch system **20** according to the first embodiment (refer to FIG. **1**) but includes a second control unit **5Q** instead of the second control unit **5**, which is a difference from the switch system **20**. In the following description, any constituent element of the switch system **20Q** according to this eighteenth embodiment, having the same function as a counterpart of the switch system **20** according to the first embodiment, will be designated by the same reference numeral as that

counterpart's, and description thereof will be omitted herein.

(212) In the switch system **20Q** according to the eighteenth embodiment, before the first control unit **4** outputs a first off signal to turn the semiconductor switch **1** off while outputting a first on signal to turn the semiconductor switch **1** on, the second control unit **5** outputs a second off signal to turn the control switch **33** off. In the switch system **20Q**, the second control unit **5Q** outputs a second on signal to turn the control switch **33** on when a first period (designated by **T1** in FIG. **24**) has passed since the output of the first off signal from the first control unit **4**. In the switch system **20Q**, the second control unit **5Q** outputs a second off signal to turn the control switch **33** off when a second period (designated by **T2** in FIG. **24**) has passed since the output of the first off signal from the first control unit **4**. The switch system **20Q** satisfies $T1 < (T2 - T1)$ and $T1 \leq T3$, where **T1** is the first period (refer to FIG. **24**), **T2** is the second period (refer to FIG. **24**), and **T3** is a third period (refer to FIG. **24**) between a point in time when the first control unit **4** has output the first off signal and a point in time when a current **I2** flowing through the voltage clamping element **2** becomes equal to zero (refer to FIG. **24**).

(213) Next, an exemplary operation of the switch system **20Q** will be described with reference to FIG. **24**. In FIG. **24**, the waveform labeled as “semiconductor switch (**Tr1**)” indicates how the first control signal supplied to the control terminal **10** of the semiconductor switch **1** (junction field effect transistor **Tr1**) changes with time. Also, in FIG. **24**, the waveform labeled as “control switch (**Q1**)” indicates how the second control signal supplied to the control switch **33** (field effect transistor **Q1**) changes with time. Furthermore, in FIG. **24**, “**Vgs**” denotes a voltage between the gate terminal **G1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor **Tr1**. Thus, “**Vgs**” denotes the voltage between the control terminal **10** and second main terminal **12** of the semiconductor switch **1**. Furthermore, in FIG. **24**, “**Vds**” denotes a voltage between the drain terminal **D1** and the source terminal **S1** with respect to the source terminal **S1** of the junction field effect transistor **Tr1**. Thus, “**Vds**” denotes the voltage between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**. The voltage clamping element **2** is connected to the semiconductor switch **1** in parallel. Thus, the voltage across the voltage clamping element **2** is the same as the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(214) FIG. **24** illustrates how the switch system **20Q** operates, for example, in a situation where a short-circuit failure occurs to the load circuit **204** at a time **t51** to cause the main current **I1** of the semiconductor switch **1** to start to increase.

(215) When the first control signal output from the first control unit **4** changes from a first on signal into a first off signal at a time **t52**, the main current **I1** flowing through the semiconductor switch **1** becomes zero and a current **I2** flows through the voltage clamping element **2**.

(216) Thereafter, when the second control signal output from the second control unit **5Q** changes from a second off signal into a second on signal at a time **t53**, the control switch **33** of the active clamping circuit **3** is turned on. In the switch system **20Q**, the breakdown voltage of the first diode **31** is smaller than the clamp voltage of the voltage clamping element **2**. Thus, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped by the active clamping circuit **3**, and part of the energy stored in the inductance component of the line **203** of the load circuit **204** is consumed by the semiconductor switch **1**.

(217) Thereafter, at a time **t54**, when a second period **T2** passes since the time **t52**, the second control signal output from the second control unit **5Q** changes from the second on signal into the second off signal. In response, a current **I2** starts to flow again through the voltage clamping element **2**, the voltage **Vds** between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** is clamped at the clamp voltage of the voltage clamping element **2**, and the absolute value of a current variation ratio ($-dI1/dt$) of the main current **I1** of the semiconductor switch **1** increases to cause the main current **I1** to decrease steeply to zero. In the switch system **20Q**, when the main current **I1** of the semiconductor switch **1** goes zero, the current **I2** of the

voltage clamping element **2** starts to decrease.

(218) When the current I_2 of the voltage clamping element **2** goes zero at a time t_{55} , the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1** becomes a constant voltage lower than the clamp voltage.

(219) The switch system **20Q** may make the current value of the main current I_1 flowing through the semiconductor switch **1** at the time t_{54} smaller than the current value of the main current I_1 flowing through the semiconductor switch **1** at the time t_{52} . This allows the switch system **20Q** to reduce the clamp voltage of the voltage clamping element **2** and thereby lower the voltage V_{ds} between the first main terminal **11** and second main terminal **12** of the semiconductor switch **1**.

(220) The switch system **20Q** according to the eighteenth embodiment may also reduce the surge voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

Nineteenth Embodiment

(221) Next, a switch system **20R** according to a nineteenth embodiment will be described with reference to FIG. **25**.

(222) The switch system **20R** according to the nineteenth embodiment is substantially the same as the switch system **20F** according to the seventh embodiment (refer to FIG. **8**) but includes a NOT circuit **7M** instead of the NOT circuit **7F**, which is a difference from the switch system **20F**. In addition, the switch system **20R** according to the nineteenth embodiment does not include the third drive circuit **13** of the switch system **20F** according to the seventh embodiment, which is another difference from the switch system **20F**. In the following description, any constituent element of the switch system **20R** according to this nineteenth embodiment, having the same function as a counterpart of the switch system **20F** according to the seventh embodiment, will be designated by the same reference numeral as that counterpart's, and description thereof will be omitted herein.

(223) The NOT circuit **7M** includes a power supply **71**, a diode **73**, and a third field effect transistor **Q3**. The power supply **71** is a DC power supply. The diode **73** is formed by connecting together the gate and source of the field effect transistor. The diode **73** is connected to the power supply **71** in series. The third field effect transistor **Q3** has a third gate, a third drain, and a third source. The third field effect transistor **Q3** is connected to the diode **73** in series. In the third field effect transistor **Q3**, the third drain thereof is connected to the diode **73** and the third source thereof is connected to the second source of the second field effect transistor **Q2** and the second main terminal **12** of the semiconductor switch **1**. The third field effect transistor **Q3** may be, for example, a junction field effect transistor. However, this is only an example and should not be construed as limiting. Alternatively, the third field effect transistor **Q3** may also be a MOSFET. The first control unit **4** and the third gate of the third field effect transistor **Q3** are connected to a signal path between the first control unit **4** and the semiconductor switch **1**.

(224) In the switch system **20R** according to the nineteenth embodiment, as well as the switch system **20Q** according to the eighteenth embodiment, before the first control unit **4** outputs a first off signal to turn the semiconductor switch **1** off while outputting a first on signal to turn the semiconductor switch **1** on, the second control unit **5E** outputs a second off signal to turn the control switch **33** off. In the switch system **20R**, the second control unit **5E** outputs a second on signal to turn the control switch **33** on when a first period has passed since the output of the first off signal from the first control unit **4**. In the switch system **20R**, the second control unit **5E** outputs a second off signal to turn the control switch **33** off when a second period (i.e., a time delay determined by the delay circuit **52**) has passed since the output of the first off signal from the first control unit **4**. The switch system **20R** satisfies $T_1 \leq (T_2 - T_1)$ and $T_1 \leq T_3$, where T_1 is the first period (refer to FIG. **24**), T_2 is the second period (refer to FIG. **24**), and T_3 is a third period (refer to FIG. **24**) between a point in time when the first control unit **4** has output the first off signal and a point in time when a current I_2 flowing through the voltage clamping element **2** becomes equal to zero.

(225) The switch system **20R** according to the nineteenth embodiment may also reduce the surge

voltage applied to the semiconductor switch **1** when the semiconductor switch **1** turns off.

(226) (Variations)

(227) Note that the first to nineteenth embodiments described above are only exemplary ones of various embodiments of the present disclosure and should not be construed as limiting. Rather, the first to nineteenth embodiments may be readily modified in various manners depending on a design choice or any other factor without departing from the scope of the present disclosure.

(228) For example, the semiconductor switch **1** does not have to be a junction field effect transistor **Tr1** but may also be, for example, a MOSFET or an insulated gate bipolar transistor (IGBT).

(229) (Aspects)

(230) The first through nineteenth embodiments and their variations described above are specific implementations of the following aspects of the present disclosure.

(231) A switch system (**20**; **20A**; **20B**; **20C**; **20D**; **20E**; **20F**; **20G**; **20J**; **20K**; **20O**; **20P**) according to a first aspect includes a semiconductor switch (**1**; **1K**), a voltage clamping element (**2**), an active clamping circuit (**3**; **3A**; **30**), a first control unit (**4**), and a second control unit (**5**; **5A**; **5D**; **5E**). The semiconductor switch (**1**; **1K**) has a control terminal (**10**), a first main terminal (**11**), and a second main terminal (**12**). The voltage clamping element (**2**) is connected to the semiconductor switch (**1**; **1K**) in parallel. The active clamping circuit (**3**; **3A**; **30**) is connected between the control terminal (**10**) and the first main terminal (**11**) of the semiconductor switch (**1**; **1K**). The first control unit (**4**) controls the semiconductor switch (**1**; **1K**). The second control unit (**5**; **5A**; **5D**; **5E**) controls the active clamping circuit (**3**; **3A**; **30**). The active clamping circuit (**3**; **3A**; **30**) includes a first diode (**31**), a second diode (**32**), and a control switch (**33**). The first diode (**31**) has a first anode and a first cathode. The first diode (**31**) causes breakdown upon application of voltage between the first main terminal (**11**) and the second main terminal (**12**) of the semiconductor switch (**1**; **1K**). The second diode (**32**) has a second anode and a second cathode. The second anode of the second diode (**32**) is connected to the first anode of the first diode (**31**). The control switch (**33**) is connected between the first anode of the first diode (**31**) and the control terminal (**10**) of the semiconductor switch (**1**). The second control unit (**5**; **5A**; **5D**; **5E**) is configured to control the control switch (**33**). A breakdown voltage of the first diode (**31**) is smaller than a clamp voltage of the voltage clamping element (**2**).

(232) The switch system (**20**; **20A**; **20B**; **20C**; **20D**; **20E**; **20F**; **20G**; **20J**; **20K**; **20O**; **20P**) according to the first aspect may reduce a surge voltage applied to the semiconductor switch (**1**; **1K**) when the semiconductor switch (**1**; **1K**) turns off.

(233) In a switch system (**20A**; **20B**; **20C**; **20D**; **20E**; **20F**; **20G**) according to a second aspect, which may be implemented in conjunction with the first aspect, the control switch (**33**) is a first field effect transistor (**Q1**) having a first gate, a first drain, and a first source. The active clamping circuit (**3A**) further includes a second field effect transistor (**Q2**). The second field effect transistor (**Q2**) has a second gate, a second drain, and a second source. The second field effect transistor (**Q2**) is connected between the first gate of the first field effect transistor (**Q1**) and the second main terminal (**12**) of the semiconductor switch (**1**; **1K**). The second control unit (**5A**; **5D**; **5E**) controls the second field effect transistor (**Q2**).

(234) The switch system (**20A**; **20B**; **20C**; **20D**; **20E**; **20F**; **20G**) according to the second aspect may turn the first field effect transistor (**Q1**) off by making the second control unit (**5A**; **5D**; **5E**) turn the second field effect transistor (**Q2**) on.

(235) A switch system (**20B**) according to a third aspect, which may be implemented in conjunction with the second aspect, further includes a delay circuit (**6**). The delay circuit (**6**) is connected between a signal path that connects the second control unit (**5A**) to the first gate of the first field effect transistor (**Q1**) and the second gate of the second field effect transistor (**Q2**).

(236) In the switch system (**20B**) according to the third aspect, the first field effect transistor (**Q1**) turns on when the second control unit (**5A**) outputs a control signal that turns the first field effect transistor (**Q1**) on, and the first field effect transistor (**Q1**) turns off when the second field effect

transistor (Q2) turns on after a time delay introduced by the delay circuit (6). Thus, the switch system (20B) according to the third aspect may determine the conductive period of the first field effect transistor (Q1) by the time delay introduced by the delay circuit (6). In addition, the switch system (20B) according to the third aspect may control the first field effect transistor (Q1) and the second field effect transistor (Q2) using a single second control unit (5A).

(237) A switch system (20C; 20D; 20E; 20F; 20G) according to a fourth aspect, which may be implemented in conjunction with the second aspect, further includes a NOT circuit (7; 7F; 7G). The NOT circuit (7; 7F; 7G) is connected between a signal path that connects the first control unit (4) to the control terminal (10) of the semiconductor switch (1; 1K) and the first gate of the first field effect transistor (Q1). The second control unit (5A; 5D; 5E) is connected to the second gate of the second field effect transistor (Q2).

(238) The switch system (20C; 20D; 20E; 20F; 20G) according to the fourth aspect may control the semiconductor switch (1; 1K) and the field effect transistor (Q1) using the first control unit (4).

(239) In a switch system (20D; 20E; 20F; 20G) according to a fifth aspect, which may be implemented in conjunction with the fourth aspect, the second control unit (5D) includes a delay circuit (52). The delay circuit (52) is connected between an output terminal of the NOT circuit (7; 7F; 7G) and the second gate of the second field effect transistor (Q2).

(240) The switch system (20D; 20E; 20F; 20G) according to the fifth aspect allows the second control unit (5D; 5E) to use a control signal supplied from the first control unit (4), thus contributing to cutting down the cost.

(241) A switch system (20E; 20F; 20G) according to a sixth aspect, which may be implemented in conjunction with the fifth aspect, further includes a first drive circuit (8). The first drive circuit (8) has a first gate resistor (Rg1). The first gate resistor (Rg1) is connected between the output terminal of the NOT circuit (7; 7F; 7G) and the first gate of the first field effect transistor (Q1). The delay circuit (52) is implemented as a second drive circuit (9) including a second gate resistor (Rg2) and a capacitor (C2). The second gate resistor (Rg2) is connected between the output terminal of the NOT circuit (7; 7F; 7G) and the second gate of the second field effect transistor (Q2). The capacitor (C2) is connected between the second gate and the second source of the second field effect transistor (Q2).

(242) The switch system (20E; 20F; 20G) according to the sixth aspect allows the delay circuit (52) to be formed by the second gate resistor (Rg2) and the capacitor (C2), thus contributing to cutting down the cost of the delay circuit (52).

(243) A switch system (20F) according to a seventh aspect, which may be implemented in conjunction with the sixth aspect, further includes a third drive circuit (13). The third drive circuit (13) is connected between the first control unit (4) and the control terminal (10) of the semiconductor switch (1; 1K). The NOT circuit (7) includes a power supply (71), a resistor (72), a third field effect transistor (Q3), and a fourth drive circuit (74). The resistor (72) is connected to the power supply (71) in series. The third field effect transistor (Q3) has a third gate, a third drain, and a third source. The third field effect transistor (Q3) is connected to the resistor (72) in series. The fourth drive circuit (74) is connected between the first control unit (4) and the third gate of the third field effect transistor (Q3).

(244) A switch system (20G) according to an eighth aspect, which may be implemented in conjunction with the sixth aspect, further includes a third drive circuit (13). The third drive circuit (13G) is connected between the first control unit (4) and the control terminal (10) of the semiconductor switch (1; 1K). The NOT circuit (7G) includes a power supply (71), a resistor (72), and a third field effect transistor (Q3). The resistor (72) is connected to the power supply (71) in series. The third field effect transistor (Q3) has a third gate, a third drain, and a third source. The third field effect transistor (Q3) is connected to the resistor (72) in series. The third drive circuit (13G) has a first path (131) to turn the semiconductor switch (1; 1K) on and a second path (132) to turn the semiconductor switch (1; 1K) off. The first path (131) includes a third diode (133) and a

third gate resistor (Rg3). The third diode (133) has a third anode and a third cathode. The third cathode of the third diode (133) is connected to the control terminal (10) of the semiconductor switch (1; 1K). The third gate resistor (Rg3) is connected to the third diode (133) in series. The second path (132) includes a resistor (134) having a smaller resistance value than the third gate resistor (Rg3). The resistor (134) is connected between the first control unit (4) and the control terminal (10) of the semiconductor switch (1; 1K). In the switch system (20G), the first field effect transistor (Q1) of the active clamping circuit (3A) turns on before the semiconductor switch (1; 1K) turns off.

(245) In the switch system (20G) according to the eighth aspect, if the semiconductor switch (1; 1K) is a GaN-based GIT, the first path (131) is used to turn the semiconductor switch (1; 1K) on, thus reducing the chances of causing a decline in the gate voltage. In addition, in this switch system (20G), the second path (132) is used to turn the semiconductor switch (1; 1K) off. This reduces the chances of the semiconductor switch (1; 1K) turning off before the control switch (33) of the active clamping circuit (3A) turns on.

(246) A switch system (20H) according to a ninth aspect includes a semiconductor switch (1; 1K), a voltage clamping element (2), an active clamping circuit (3H), a first control unit (4), and a second control unit (5H). The semiconductor switch (1; 1K) has a control terminal (10), a first main terminal (11), and a second main terminal (12). The voltage clamping element (2) is connected to the semiconductor switch (1; 1K) in parallel. The active clamping circuit (3H) is connected between the control terminal (10) and the first main terminal (11) of the semiconductor switch (1; 1K). The first control unit (4) controls the semiconductor switch (1; 1K). The second control unit (5H) controls the active clamping circuit (3H). The active clamping circuit (3H) includes a first diode (31), a second diode (32), and a control switch (33H). The first diode (31) has a first anode and a first cathode. The first diode (31) causes breakdown upon application of voltage between the first main terminal (11) and the second main terminal (12) of the semiconductor switch (1; 1K). The second diode (32) has a second anode and a second cathode. The second anode of the second diode (32) is connected to the first anode of the first diode (31). The control switch (33H) is connected between the first anode of the first diode (31) and the control terminal (10) of the semiconductor switch (1; 1K). The control switch (33H) is a pnp transistor (Qp1) having a base, a collector, and an emitter. The emitter of the pnp transistor (Qp1) is connected to the second cathode of the second diode (32). The collector of the pnp transistor (Qp1) is connected to the control terminal (10) of the semiconductor switch (1; 1K). The second control unit (5H) includes a first resistor (R1), a second resistor (R2), a field effect transistor (Q4), and a delay circuit (54). The first resistor (R1) is connected between the base and the emitter of the pnp transistor (Qp1). The second resistor (R2) is connected to the base of the pnp transistor (Qp1) and connected to the first resistor (R1) in series. The field effect transistor (Q4) has a gate, a drain, and a source. The field effect transistor (Q4) is connected between the second resistor (R2) and the second main terminal (12) of the semiconductor switch (1; 1K). The delay circuit (54) is connected between the first control unit (4) and the gate of the field effect transistor (Q4). The delay circuit (54) includes a gate resistor (Rg4) and a capacitor (C4). The gate resistor (Rg4) is connected between the first control unit (4) and the gate of the field effect transistor (Q4). The capacitor (C4) is connected between the gate and the source of the field effect transistor (Q4). A breakdown voltage of the first diode (31) is smaller than a clamp voltage of the voltage clamping element (2).

(247) The switch system (20H) according to the ninth aspect may reduce a surge voltage applied to the semiconductor switch (1; 1K) when the semiconductor switch (1; 1K) turns off.

(248) A switch system (20I) according to a tenth aspect includes a semiconductor switch (1; 1K), a voltage clamping element (2I), an active clamping circuit (3I), a first control unit (4), and a second control unit (5I). The semiconductor switch (1; 1K) has a control terminal (10), a first main terminal (11), and a second main terminal (12). The voltage clamping element (2I) is connected to the semiconductor switch (1; 1K) in parallel. The active clamping circuit (3I) is connected between

the control terminal (10) and the first main terminal (11) of the semiconductor switch (1; 1K). The first control unit (4) controls the semiconductor switch (1; 1K). The second control unit (5I) controls the active clamping circuit (3I). The active clamping circuit (3I) includes a first diode (31), a second diode (32), and a control switch (33). The first diode (31) has a first anode and a first cathode. The first diode (31) causes breakdown upon application of voltage between the first main terminal (11) and the second main terminal (12) of the semiconductor switch (1; 1K). The second diode (32) has a second anode and a second cathode. The second cathode of the second diode (32) is connected to the first cathode of the first diode (31). The control switch (33) is connected between the first anode of the first diode (31) and the control terminal (10) of the semiconductor switch (1; 1K). The control switch (33) is a field effect transistor (Q1) having a gate, a drain, and a source. The drain of the field effect transistor (Q1) is connected to the first anode of the first diode (31). The source of the field effect transistor (Q1) is connected to the control terminal (10) of the semiconductor switch (1; 1K). The second control unit (5I) includes a capacitor (55) and a Zener diode (56). The capacitor (55) has a first terminal and a second terminal. The first terminal of the capacitor (55) is connected to the gate of the field effect transistor (Q1). The second terminal of the capacitor (55) is connected to the first cathode of the first diode (31) and the second cathode of the second diode (32). The Zener diode (56) has a third anode and a third cathode. The third cathode of the Zener diode (56) is connected to the first terminal of the capacitor (55) and the gate of the field effect transistor (Q1). The third anode of the Zener diode (56) is connected to the source of the field effect transistor (Q1).

(249) The switch system (20I) according to the tenth aspect may reduce a surge voltage applied to the semiconductor switch (1; 1K) when the semiconductor switch (1; 1K) turns off.

(250) A switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J) according to an eleventh aspect, which may be implemented in conjunction with any one of the first to tenth aspects, further includes a detection circuit (14) and a switch element (15). The detection circuit (14) is connected between the control terminal (10) and the second main terminal (12) of the semiconductor switch (1). The detection circuit (14) detects a voltage between the control terminal (10) and the second main terminal (12) of the semiconductor switch (1; 1K). The switch element (15) is connected between the control terminal (10) and the second main terminal (12) of the semiconductor switch (1; 1K). In the switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J), the switch element (15) turns on when the voltage detected by the detection circuit (14) exceeds a threshold voltage (V_{th}).

(251) In the switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J) according to the eleventh aspect, when the switch element (15) turns on, the semiconductor switch (1; 1K) turns off. This reduces the chances of causing deterioration in the characteristic of the semiconductor switch (1; 1K) due to an increase in voltage between the control terminal (10) and second main terminal (12) of the semiconductor switch (1; 1K).

(252) In a switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J) according to a twelfth aspect, which may be implemented in conjunction with the eleventh aspect, the semiconductor switch (1) is a junction field effect transistor (Tr1). With respect to a gate voltage (V_{gs}) of the junction field effect transistor (Tr1), the threshold voltage (V_{th}) is larger than a gate voltage (V_{g1}) of the junction field effect transistor (Tr1) in on state and smaller than an output voltage of a DC power supply included in the first control unit (4).

(253) The switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J) according to the twelfth aspect may reduce deterioration in the characteristics of the semiconductor switch (1) due to an increase in the gate voltage (V_{gs}) of the junction field effect transistor (Tr1) while the active clamping circuit (3; 3A) is performing a clamping operation.

(254) In a switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J; 20K) according to a thirteenth aspect, which may be implemented in conjunction with any one of the first to twelfth aspects, the semiconductor switch (1K) is made up of a plurality of field effect transistors (Tr2) that

are connected in parallel. Each of the plurality of field effect transistors (Tr2) has a gate, a drain, and a source. The first main terminal (11) of the semiconductor switch (1K) is connected to respective drains of the plurality of field effect transistors (Tr2). The second main terminal (12) of the semiconductor switch (1K) is connected to respective sources of the plurality of field effect transistors (Tr2). The switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J; 20K) further includes a plurality of gate resistors (R20). The plurality of gate resistors (R20) are associated one to one with respective gates and connected between the first control unit (4) and the respective gates. The plurality of gate transistors (R20) are connected to the active clamping circuit (3; 3A).

(255) The switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20I; 20J; 20K) according to the thirteenth aspect allows the semiconductor switch (1K) to have a permissible current with an increased current value.

(256) In a switch system (20L) according to a fourteenth aspect, which may be implemented in conjunction with any one of the first to thirteenth aspects, the semiconductor switch (1L) includes a first switch element (101) and a second switch element (102). The first switch element (101) has a first gate terminal (G11), a first drain terminal (D11), and a first source terminal (S11). The second switch element (102) has a second gate terminal (G12), a second drain terminal (D12), and a second source terminal (S12). The second switch element (102) is connected to the first switch element (101) in series. In the semiconductor switch (1L), the first drain terminal (D11) of the first switch element (101) and the second drain terminal (D12) of the second switch element (102) are connected to each other. The semiconductor switch (1L) includes two sets, each consisting of the control terminal, the first main terminal, and the second main terminal. In one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal (G11), the second source terminal (S12), and the first source terminal (S11), respectively. In the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal (G12), the first source terminal (S11), and the second source terminal (S12), respectively. The switch system (20L) includes two active clamping circuits (3A), each of which is the active clamping circuit (3A). In the switch system (20L), one of the two active clamping circuits (3A) is connected between the first gate terminal (G11) and the second source terminal (S12), and the other active clamping circuit (3A) is connected between the second gate terminal (G12) and the first source terminal (S11).

(257) The switch system (20L) according to the fourteenth aspect is applicable to a load circuit (204) including an AC power supply.

(258) A switch system (20M) according to a fifteenth aspect, which may be implemented in conjunction with any one of the first to thirteenth aspects, includes two semiconductor switches (1), each of which is the semiconductor switch (1), two voltage clamping elements (2), each of which is the voltage clamping element (2), and two active clamping circuits (3A), each of which is the active clamping circuit (3A). In each of the two semiconductor switches (1), the control terminal (10), the first main terminal (11), and the second main terminal (12) thereof are a gate terminal (G1), a drain terminal (D1), and a source terminal (S1), respectively. In the switch system (20M), the two semiconductor switches (1) are connected in series by connecting respective source terminals (S1) thereof together. The two voltage clamping elements (2) are associated one to one with the two semiconductor switches (1), respectively. Each of the two voltage clamping elements (2) is connected to an associated one of the two semiconductor switches (1) in parallel. The two active clamping circuits (3A) are associated one to one with the two semiconductor switches (1), respectively. Each of the two active clamping circuits (3A) is connected between the gate terminal (G1) and the drain terminal (D1) of an associated one of the two semiconductor switches (1).

(259) The switch system (20M) according to the fifteenth aspect is applicable to a load circuit (204) including an AC power supply.

(260) In a switch system (20N) according to a sixteenth aspect, which may be implemented in

conjunction with any one of the first to thirteenth aspects, the semiconductor switch (1N) is a dual-gate bidirectional switch having a first gate terminal (G11), a first source terminal (S11) corresponding to the first gate terminal (G11), a second gate terminal (G12), and a second source terminal (S12) corresponding to the second gate terminal (G12). The semiconductor switch (1N) includes two sets, each consisting of the control terminal, the first main terminal, and the second main terminal. In one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal (G11), the second source terminal (S12), and the first source terminal (S11), respectively. In the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal (G12), the first source terminal (S11), and the second source terminal (S12), respectively. The switch system (20N) includes two active clamping circuits (3A), each of which is the active clamping circuit (3A). In the switch system (20N), one of the two active clamping circuits (3A) is connected between the first gate terminal (G11) and the second source terminal (S12), and the other active clamping circuit (3A) is connected between the second gate terminal (G12) and the first source terminal (S11).

(261) The switch system (20N) according to the sixteenth aspect is applicable to a load circuit (204) including an AC power supply.

(262) A switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20J; 20K; 20L; 20M; 20N; 20O; 20P; 20Q; 20R) according to a seventeenth aspect, which may be implemented in conjunction with any one of the first to sixteenth aspects, further includes a current detection unit (16). The current detection unit (16) detects a current flowing through the second main terminal (12) of the semiconductor switch (1; 1K; 1L; 1N). The second control unit (5; 5A; 5H; 5E) changes a conductive period of the control switch (33; 33H) according to output of the current detection unit (16).

(263) The switch system (20; 20A; 20B; 20C; 20D; 20E; 20F; 20G; 20H; 20J; 20K; 20L; 20M; 20N; 20O; 20P; 20Q; 20R) according to the seventeenth aspect may reduce the chances of allowing an overcurrent to flow through the voltage clamping element (2) to cause deterioration in the performance of the voltage clamping element (2).

(264) A switch system (20P) according to an eighteenth aspect, which may be implemented in conjunction with the second aspect, further includes a current detection unit (16P). The second control unit (5P) includes a third field effect transistor (Q13). The current detection unit (16P) detects a current flowing through the second main terminal (12) of the semiconductor switch (1). The third field effect transistor (Q13) has a third gate, a third drain, and a third source. The current detection unit (16P) includes a detecting resistor (161) and an integration circuit (162). The detecting resistor (161) is connected to the second main terminal (12) of the semiconductor switch (1) and connected to the semiconductor switch (1) in series. The integration circuit (162) integrates voltage across the detecting resistor (161). In the switch system (20P), the second gate of the second field effect transistor (Q2) is connected to an output terminal of the integration circuit (162). In the switch system (20P), the third drain of the third field effect transistor (Q13) is connected to the output terminal of the integration circuit (162). In the switch system (20P), the third gate of the third field effect transistor (Q13) is connected to a signal path between the first control unit (4) and the control terminal (10) of the semiconductor switch (1).

(265) In the switch system (20P) according to the eighteenth aspect, the timing when the second field effect transistor (Q2) turns on varies according to the amount of current flowing through the second main terminal (12) of the semiconductor switch (1), thus changing the duration for which the first field effect transistor (Q1) is energized.

(266) In a switch system (20) according to a nineteenth aspect, which may be implemented in conjunction with the first aspect, the second control unit (5) outputs a second on signal to turn the control switch (33) on, while the first control unit (4) is outputting a first on signal to turn the semiconductor switch (1) on before outputting a first off signal to turn the semiconductor switch (1) off. In the switch system (20), the second control unit (5) outputs a second off signal to turn the

control switch (33) off when a predetermined period (TO) has passed since output of the first off signal from the first control unit (4).

(267) In a switch system (20Q; 20R) according to a twentieth aspect, which may be implemented in conjunction with the first aspect, the second control unit (5Q; 5E) outputs a second off signal to turn the control switch (33) off, while the first control unit (4) is outputting a first on signal to turn the semiconductor switch (1) on before outputting a first off signal to turn the semiconductor switch (1) off. In the switch system (20Q; 20R), the second control unit (5) outputs a second on signal to turn the control switch (33) on when a first period has passed since output of the first off signal from the first control unit (4). In the switch system (20Q; 20R), the second control unit (5Q; 5E) outputs a second off signal to turn the control switch (33) off when a second period has passed since the output of the first off signal from the first control unit (4). The switch system (20Q; 20R) satisfies $T1 \leq (T2 - T1)$ and $T1 \leq T3$, where T1 is the first period, T2 is the second period, and T3 is a third period between a point in time when the first control unit 4 has output the first off signal and a point in time when a current (I2) flowing through the voltage clamping element (2) becomes equal to zero.

REFERENCE SIGNS LIST

(268) 1, 1K, 1L, 1N Semiconductor Switch 10 Control Terminal 11 First Main Terminal 12 Second Main Terminal 101 First Switch Element 102 Second Switch Element 2, 2i Voltage Clamping Element 3, 3A, 3H, 3I, 30 Active Clamping Circuit 31 First Diode 32 Second Diode 33, 33H Control Switch 4 First Control Unit 5, 5A, 5D, 5E, 5H, 5I, 5Q Second Control Unit 52 Delay Circuit 54 Delay Circuit 55 Capacitor 56 Zener Diode 6 Delay Circuit 7, 7F, 7G, 7M Not Circuit 71 Power Supply 72 Resistor 74 Fourth Drive Circuit 8 First Drive Circuit 9 Second Drive Circuit 13, 13G Third Drive Circuit 131 First Path 132 Second Path 133 Third Diode 134 Resistor 14 Detection Circuit 15 Switch Element 16 Current Detection Unit 16P Current Detection Unit 161 Detecting Resistor 162 Integration Circuit 20, 20A, 20B, 20C, 20D, 20E, 20F, 20G, 20H, 20I, 20J, 20K, 20L, 20M, 20N, 20O, 20P, 20Q, 20R Switch System C2 Capacitor C4 Capacitor D1 Drain Terminal D11 First Drain Terminal D12 Second Drain Terminal G1 Gate Terminal G11 First Gate Terminal G12 Second Gate Terminal Q1 Field Effect Transistor (First Field Effect Transistor) Q2 Second Field Effect Transistor Q3 Third Field Effect Transistor Q4 Field Effect Transistor Q13 Third Field Effect Transistor Qp1 pnp Transistor R1 First Resistor R2 Second Resistor Rg1 First Gate Resistor Rg2 Second Gate Resistor Rg3 Third Gate Resistor Rg4 Gate Resistor R20 Gate Resistor S1 Source Terminal Tr1 Junction Field Effect Transistor Tr2 Field Effect Transistor

Claims

1. A switch system comprising: a semiconductor switch having a control terminal, a first main terminal, and a second main terminal; a voltage clamping element connected to the semiconductor switch in parallel; an active clamping circuit connected between the control terminal and the first main terminal of the semiconductor switch; a first control unit configured to control the semiconductor switch; a second control unit configured to control the active clamping circuit; and a NOT circuit, the active clamping circuit including: a first diode having a first anode and a first cathode and causing breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch; a second diode having a second anode and a second cathode, the second anode of the second diode being connected to the first anode of the first diode; and a control switch connected between the first anode of the first diode and the control terminal of the semiconductor switch, the second control unit being configured to control the control switch, a breakdown voltage of the first diode being smaller than a clamp voltage of the voltage clamping element, wherein the control switch is a first field effect transistor having a first gate, a first drain, and a first source, the active clamping circuit further includes a second field effect transistor having a second gate, a second drain, and a second source and connected between

the first gate of the first field effect transistor and the second main terminal of the semiconductor switch, the second control unit is configured to control the second field effect transistor, the NOT circuit is connected between a signal path that connects the first control unit to the control terminal of the semiconductor switch and the first gate of the first field effect transistor, and the second control unit is connected to the second gate of the second field effect transistor.

2. The switch system of claim 1, wherein the second control unit includes a delay circuit, and the delay circuit is connected between an output terminal of the NOT circuit and the second gate of the second field effect transistor.

3. The switch system of claim 2, further comprising a first drive circuit having a first gate resistor connected between the output terminal of the NOT circuit and the first gate of the first field effect transistor, wherein the delay circuit is implemented as a second drive circuit, the second drive circuit including: a second gate resistor connected between the output terminal of the NOT circuit and the second gate of the second field effect transistor; and a capacitor connected between the second gate and the second source of the second field effect transistor.

4. The switch system of claim 3, further comprising a third drive circuit connected between the first control unit and the control terminal of the semiconductor switch, wherein the NOT circuit includes: a power supply; a resistor connected to the power supply in series; a third field effect transistor having a third gate, a third drain, and a third source and connected to the resistor in series; and a fourth drive circuit connected between the first control unit and the third gate of the third field effect transistor.

5. The switch system of claim 3, further comprising a third drive circuit connected between the first control unit and the control terminal of the semiconductor switch, wherein the NOT circuit includes: a power supply; a resistor connected to the power supply in series; and a third field effect transistor having a third gate, a third drain, and a third source and connected to the resistor in series, the third drive circuit has: a first path configured to turn the semiconductor switch on; and a second path configured to turn the semiconductor switch off, the first path includes: a third diode having a third anode and a third cathode, the third cathode of the third diode being connected to the control terminal of the semiconductor switch; and a third gate resistor connected to the third diode in series, the second path includes a resistor connected between the first control unit and the control terminal of the semiconductor switch and having a smaller resistance value than the third gate resistor, and the first field effect transistor of the active clamping circuit being configured to turn on before the semiconductor switch turns off.

6. A switch system comprising: a semiconductor switch having a control terminal, a first main terminal, and a second main terminal; a voltage clamping element connected to the semiconductor switch in parallel; an active clamping circuit connected between the control terminal and the first main terminal of the semiconductor switch; a first control unit configured to control the semiconductor switch; a second control unit configured to control the active clamping circuit; a detection circuit connected between the control terminal and the second main terminal of the semiconductor switch to detect a voltage between the control terminal and the second main terminal of the semiconductor switch; and a switch element connected between the control terminal and the second main terminal of the semiconductor switch, the active clamping circuit including: a first diode having a first anode and a first cathode and causing breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch; a second diode having a second anode and a second cathode, the second anode of the second diode being connected to the first anode of the first diode; and a control switch connected between the first anode of the first diode and the control terminal of the semiconductor switch, the second control unit being configured to control the control switch, a breakdown voltage of the first diode being smaller than a clamp voltage of the voltage clamping element, wherein the switch element is configured to turn on when the voltage detected by the detection circuit exceeds a threshold voltage.

7. The switch system of claim 6, wherein the semiconductor switch is a junction field effect transistor, with respect to a gate voltage of the junction field effect transistor, the threshold voltage is larger than a gate voltage of the junction field effect transistor in on state and smaller than an output voltage of a DC power supply included in the first control unit.

8. The switch system of claim 1, wherein the semiconductor switch is made up of a plurality of field effect transistors that are connected in parallel, each of the plurality of field effect transistors has a gate, a drain, and a source, the first main terminal of the semiconductor switch is connected to respective drains of the plurality of field effect transistors, the second main terminal of the semiconductor switch is connected to respective sources of the plurality of field effect transistors, the switch system further includes a plurality of gate resistors associated one to one with respective gates of the plurality of field effect transistors and connected between the first control unit and the respective gates, and the plurality of gate transistors are connected to the active clamping circuit.

9. The switch system of claim 1, comprising: a first switch element having a first gate terminal, a first drain terminal, and a first source terminal; and a second switch element having a second gate terminal, a second drain terminal, and a second source terminal and connected to the first switch element in series, wherein in the semiconductor switch, the first drain terminal of the first switch element and the second drain terminal of the second switch element are connected to each other, the semiconductor switch includes two sets, each consisting of the control terminal, the first main terminal, and the second main terminal, in one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal, the second source terminal, and the first source terminal, respectively, in the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal, the first source terminal, and the second source terminal, respectively, the switch system includes two active clamping circuits, each of which is the active clamping circuit, and one of the two active clamping circuits is connected between the first gate terminal and the second source terminal and the other of the two active clamping circuits is connected between the second gate terminal and the first source terminal.

10. The switch system of claim 1, comprising: two semiconductor switches, each of which is the semiconductor switch; two voltage clamping elements, each of which is the voltage clamping element; and two active clamping circuits, each of which is the active clamping circuit, wherein in each of the two semiconductor switches, the control terminal, the first main terminal, and the second main terminal thereof are a gate terminal, a drain terminal, and a source terminal, respectively, the two semiconductor switches are connected in series by connecting respective source terminals thereof together, the two voltage clamping elements are associated one to one with the two semiconductor switches, respectively, each of the two voltage clamping elements being connected to an associated one of the two semiconductor switches in parallel, and the two active clamping circuits are associated one to one with the two semiconductor switches, respectively, each of the two active clamping circuits being connected between the gate terminal and the drain terminal of an associated one of the two semiconductor switches.

11. The switch system of claim 1, wherein the semiconductor switch is a dual-gate bidirectional switch having a first gate terminal, a first source terminal corresponding to the first gate terminal, a second gate terminal, and a second source terminal corresponding to the second gate terminal, the semiconductor switch includes two sets, each consisting of the control terminal, the first main terminal, and the second main terminal, in one of the two sets, the control terminal, the first main terminal, and the second main terminal are the first gate terminal, the second source terminal, and the first source terminal, respectively, in the other of the two sets, the control terminal, the first main terminal, and the second main terminal are the second gate terminal, the first source terminal, and the second source terminal, respectively, the switch system includes two active clamping circuits, each of which is the active clamping circuit, and one of the two active clamping circuits is connected between the first gate terminal and the second source terminal and the other of the two

active clamping circuits is connected between the second gate terminal and the first source terminal.

12. A switch system comprising: a semiconductor switch having a control terminal, a first main terminal, and a second main terminal; a voltage clamping element connected to the semiconductor switch in parallel; an active clamping circuit connected between the control terminal and the first main terminal of the semiconductor switch; a first control unit configured to control the semiconductor switch; a second control unit configured to control the active clamping circuit; and a current detection unit configured to detect a current flowing through the second main terminal of the semiconductor switch, the active clamping circuit including: a first diode having a first anode and a first cathode and causing breakdown upon application of voltage between the first main terminal and the second main terminal of the semiconductor switch; a second diode having a second anode and a second cathode, the second anode of the second diode being connected to the first anode of the first diode; and a control switch connected between the first anode of the first diode and the control terminal of the semiconductor switch, the second control unit being configured to control the control switch, a breakdown voltage of the first diode being smaller than a clamp voltage of the voltage clamping element, wherein the second control unit is configured to change a conductive period of the control switch according to output of the current detection unit.
